

# Joint optimization of train scheduling and rolling stock circulation planning with passenger flow control on tidal overcrowded metro lines

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## ABSTRACT

This study proposes a joint optimization method for train scheduling and rolling stock circulation planning with the consideration of passenger flow control strategy on a tidal oversaturated metro line, in which different types of rolling stocks with various loading capacities are put into operations to satisfy the uneven passenger demand in different periods (e.g., peak hours and off-peak hours). To characterize the problem mathematically, a mixed-integer nonlinear programming model is formulated to minimize the passenger waiting time and operating costs of the metro system simultaneously. This model is further reformulated equivalently into a mixed-integer linear programming model via the linearization method. An effective heuristic algorithm based on the tabu search and CPLEX solver is designed to find high-quality solutions for the proposed problem. Finally, two sets of numerical examples, including a small-scale example and a large-scale example based on the Beijing metro Batong line, are conducted to validate the performance of the proposed methods. The experimental results demonstrate that scheduling multiple types of rolling stocks can effectively reduce the transportation costs and satisfy passenger demand in different periods.

## 1. Introduction

The metro system plays a crucial role in urban public transportation owing to its large capacity, environmental friendliness, safety and punctuality, which attracts a substantial number of commuters in some megacities. To coordinate with the dynamic passenger demand, dispatchers in general use different operation strategies in peak and off-peak hours. Specifically, high-density services are provided in peak hours for vast passenger demand with the maximal transportation capacity, while the train headway is often enlarged in the off-peak periods to reduce the operating costs. Nevertheless, enlarging the headway during off-peak hours unavoidably increases the passenger waiting time, potentially leading to the decrease of service quality. In reality, besides planning operations from the perspective of timetabling, multiple types of rolling stocks (or trains) can also be utilized in different time periods to well match the passenger flow and train services, especially for metro lines with uneven passenger distributions. For instance, we can use the high-capacity and low-capacity rolling stocks during peak and off-peak hours, respectively. Since operating low-capacity rolling stocks can provide relatively high-density services for passengers with less costs, this type of operation strategy can better match the low passenger demand and train flow, and further reduce the transportation costs. In reality, this mode has already been applied in Beijing Daxing International Airport Express. In detail, unlike only operating one type of rolling stocks with

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8 carriages previously, some small rolling stocks with only 4 carriages has also been putting into operations in order to reduce the redundant capacities when there are few passenger demand. Note that, although this operation mode typically brings benefits to the metro system, it certainly increases the complexity of the operation planning process (e.g., the same type of rolling stocks should be used for operating any two connected services). In this case, developing an effective method for generating high-quality operation schemes is an important issue for realistic operations with different types of rolling stocks.

According to Lusby et al. (2011), the train scheduling (i.e., timetabling) and rolling stock planning are two decision-making components in planning the operation schemes. In general, a train schedule is firstly optimized according to the spatial and temporal distribution of passenger demand; afterwards, the rolling stocks are assigned to undertake all the scheduled services with the aim of minimizing the operating costs. In this process, the coupling relationship between these two components should be taken into account, in order to keep the feasibility of the generated operation plan. If the train schedule is not consistent with the available rolling stocks, it should be rescheduled by further considering the rolling stock constraints. Since this sequential decision might involve a tediously repeated searching process, jointly optimizing the train schedule and rolling stock circulation plan will be an expected method to generate a system-wise optimal operation plan, especially with different types of rolling stocks in operations.

For many metro lines that connect suburb areas and downtown areas, a large volume of commuters often travel to their offices from downtown areas in the morning peak hours and return home after work in the evening peak hours, leading to significant tidal and unbalance characteristics (i.e., passengers during peak hours suffer from overcrowding situations along one direction while the demand along the other direction or during the other off-peak hours is small). In this case, the operation scheme with multiple types of rolling stocks is a much promising way to reduce the operating costs with the guaranteed service quality, by dispatching more high-capacity rolling stocks during the peak hours and low-capacity rolling stocks during the other off-peak hours. In addition, on the over-saturated metro lines, imposing passenger flow control turns out to be an effective measure to avoid over crowdedness and operation risks on platforms, and the passenger flow control rules should also be made in consistence with the operation schemes (e.g., Xu et al., 2016; Shi et al., 2018). Thus, in order to further improve operation efficiency with the guaranteed traffic safety, it is desirable to design the jointly optimal operation scheme associated with the train schedule, rolling stock circulation plan and passenger flow control rule with different types of rolling stocks, by considering a series of coupling relations for different components.

### 1.1. Literature review

In the metro system, the operation planning involves a very complex process since various operation schemes, e.g., the train schedule, rolling stock circulation plan, and passenger flow control strategy, must be well coordinated. In recent years, these problems have attracted tremendous attentions from numerous practitioners and researchers. Note that the train schedule provides fundamental guidance for train operations. We next review some related studies from the following three perspectives.

#### (1) Train scheduling with dynamic passenger demand.

Train scheduling, which focuses on designing departure and arrival times of each service at its visited stations, plays a crucial role for train operations. In the literature, a variety of studies have considered the train scheduling problem for metro systems and regular railways (e.g., Shafia et al., 2012; Schmcker et al., 2011; Upadhyay and Bolia, 2014; Cacchiani et al., 2016; Canca et al., 2018; D'Ariano et al., 2007; Arbex and Cunha, 2015; Sparing and Goverde, 2017). In contrast to regular railways, the passenger demand in metro systems always has notably dynamic and random characteristics. Accordingly, the train schedule is required to coordinate with the spatial-temporal distribution of passenger demand to improve the service quality. This problem is referred to as service-oriented train scheduling problem (e.g., Robenek et al., 2018, 2016; Hassannayebi et al., 2019; Huang et al., 2016; Canca et al., 2014; Albrecht, 2019; Yin et al., 2017; Barrera et al., 2014; Sun et al., 2014) and is usually modeled as mixed-integer programming models with linear/nonlinear formulations. For instance, Mo et al. (2019b) considered different types of trains (i.e., trains with maximum capacities of 1800 persons and 1200 persons) and proposed a nonlinear integer programming model to minimize the passenger waiting time and energy costs. Then, a modified tabu search algorithm (MTS) was designed to solve the model. Considering female passenger demand, Qi et al. (2019) considered two types of carriages (i.e., general passenger cars and women-only passenger cars) for each train, and formulated a simulation-based model to minimize the total waiting time of passengers and the number of female passengers boarding general cars. Niu et al. (2015) proposed an algorithmically nonlinear mixed-integer programming model with the pre-specified skip-stop patterns to minimize the total passenger waiting time, which could be solved by the AlphaECP/GAM solver. Wang et al. (2015) formulated an event-driven train scheduling model to satisfy passenger demand in an urban metro network with the consideration of passenger transfer behaviors. Niu and Zhou (2013) developed a nonlinear integer programming model to minimize the total number of waiting passengers and weighted remaining passengers under oversaturated conditions, and a genetic algorithm was designed to solve the near-optimal solutions. Additionally, considering an oversaturated and time-dependent passenger flow, Gao et al. (2016) proposed an optimization model to reschedule a metro line in case of a disruption.

#### (2) Train scheduling with passenger flow control.

In addition, on some busy lines, the large passenger flow might lead to oversaturation at some stations, causing potential accident risks for metro operations. To reduce the operation risk and further improve the service level, a variety of studies have focused on train schedule and passenger flow control optimization over a single metro line or metro network (e.g., Davidich et al., 2013; Fernandez et al., 2015; Xu et al., 2016; Yang et al., 2017). For example, Li et al. (2017) applied a coupled state-space model to jointly optimize train regulation and passenger flow control to eliminate the delays caused by disturbances. For an oversaturated metro line, Shi et al. (2018) first formulated an integer linear programming model to collaboratively optimize the train schedule and accurate passenger flow control strategy with the consideration of minimizing the total passenger waiting time at all the involved

stations. Furthermore, they extended the model to a metro network by considering operational risk thresholds (Shi et al., 2019). To improve the accuracy of passenger flow control strategy at a single station, Li et al. (2019) proposed a dynamic radial basis function (RBF) neural network to forecast passenger volumes; then, the predicted passenger demand was used as the input data for passenger flow control strategy at a single station. Yuan et al. (2020) developed a passenger flow control model on the network to minimize the total passenger waiting time, where the control stations, control durations and control intensities were determined in the model.

### (3) Train scheduling with rolling stock circulation planning

Except providing high-quality services for the passengers, a cost-effective rolling stock circulation plan is also preferable from the perspective of managers. In the literature, many models and algorithms related to rolling stock circulation planning have been proposed, e.g., Alfieri et al. (2006), Fioole et al. (2006), Cadarso and Marín (2012), Michaelis and Schöbel (2009), Wang et al. (2018), Wagenaar et al. (2017b), Tréfond et al. (2017). Practically, since the rolling stock circulation plan is closely related to the train schedule, the collaborative optimization of these two parts is also an important topic in the literature (Peeters and Kroon, 2008; Wagenaar et al., 2017a; Schöbel, 2017). For example, considering that the train schedule is limited by the stopping patterns, Veulenturf et al. (2017) designed a heuristic approach to solve the rolling stock rescheduling problem with dynamic passenger demand. Wang et al. (2017a) proposed a two-stage optimization model to optimize the train schedule and rolling stock circulation plan with consideration of passenger demand. Then, based on the given service patterns, they further developed a mixed-integer linear programming model to jointly optimize the train schedule and rolling stock circulation plan (Wang et al., 2017b). Yue et al. (2017) proposed a bi-level programming model for an integrated train scheduling and rolling stock circulation planning problem, where the train schedule was optimized in the upper-level model and the rolling stock circulation plan was generated in the lower-level model. Liu et al. (2020) formulated a mixed-integer linear programming model to collaboratively optimize the train schedule and rolling stock circulation plan with consideration of passenger flow control strategy, in which only the same type of rolling stocks are involved.

## 1.2. The proposed methods

As mentioned above, a large number of studies in the literature have explored the collaborative optimization methods related to the train scheduling problem for metro systems. To our knowledge, however, there is still no study that specifically combines train scheduling, rolling stock circulation planning and passenger flow control strategy with consideration of different types of rolling stocks. As this topic is a promising approach for achieving system-wise optimal operation strategies, this study in particular focuses on investigating this problem. For comparison, the detailed features of some closely related studies are listed in Table 1, including decision variables, operation strategies (OS), rolling stock types (RST), objective and solution methods. As shown, only one study that focuses on train scheduling with both rolling stock circulation and passenger flow control is found in the current literature; meanwhile, the majority of studies only consider the same type of rolling stocks, which typically overlook the benefits of operating different types of rolling stocks. With these concerns, this study proposes a joint optimization model for these components from a new perspective. The detailed contributions of this study are summarized as follows.

(i) As a novel idea, this study first proposes a joint optimization method of the train schedule, passenger flow control strategy and rolling stock circulation plan with consideration of different types of rolling stocks. Compared with the traditional studies that only involve the same type of rolling stocks, this method can better satisfy the passenger demand over different periods, and further reduce the operating costs simultaneously for metro systems.

(ii) A mixed-integer nonlinear programming model is developed for the studied problem, where the objective function aims to minimize the passenger waiting time and operating costs. Then, this model is equivalently transformed into a mixed-integer linear programming model by introducing linearization techniques. An effective heuristic algorithm based on the tabu search algorithm and CPLEX solver is developed to find the near-optimal solutions.

(iii) Two sets of numerical experiments, including a small-scale experiment and a large-scale experiment on the Beijing metro Batong line, are conducted to verify the effectiveness and efficiency of the proposed methods. The computational results validate that using multiple types of rolling stocks can substantially benefit the metro system from the perspectives of both service quality and operating costs, in comparison to the case with only the same type of rolling stocks.

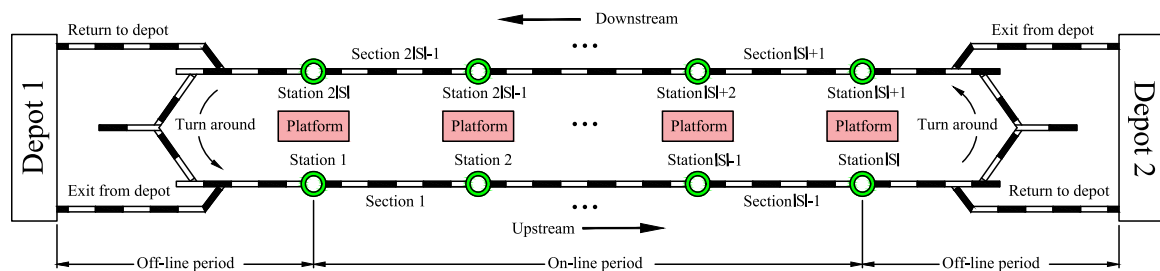
The rest of this study is organized as follows. Section 2 provides the detailed problem statements and assumptions. In Section 3, a nonlinear optimization model is formulated to characterize the problem of interest, which is finally reformulated as a mixed-integer linear programming model by some linearization techniques. Section 4 develops a heuristic algorithm base on the tabu search algorithm and CPLEX solver to effectively search the jointly optimized operation strategies. In Section 5, several experiments based on two examples are conducted to demonstrate the performance of the proposed approaches. Section 6 presents some conclusions and directions for future research.

## 2. Problem statements

This study considers a bidirectional urban rail transit line with  $2|S|$  stations and two depots, as shown in Fig. 1. The sets of upstream stations and downstream stations are denoted as  $\{1, 2, \dots, |S|\}$  and  $\{|S| + 1, |S| + 2, \dots, 2|S|\}$ , respectively. The depots are located at both ends of the rail corridor, where stations 1 and  $2|S|$  connect with depot 1 and stations  $|S|$  and  $|S| + 1$  link with depot 2. Each train runs on this metro line with a “semicircle” operation in different directions. For instance, a train departing from depot 1 is dispatched to execute service from stations 1 to  $|S|$  along the upstream direction. Upon arriving at station  $|S|$ , this train has

**Table 1**  
Feature comparison of different closely related studies.

| Publications       | Decision variables                                                   | OS                  | RST | Objective                                                            | Solution algorithm                                         |
|--------------------|----------------------------------------------------------------------|---------------------|-----|----------------------------------------------------------------------|------------------------------------------------------------|
| Niu et al. (2015)  | Train schedule                                                       | –                   | No  | Passenger waiting time                                               | GAMS                                                       |
| Yin et al. (2016)  | Train schedule                                                       | Cycle               | No  | Train delay, passenger travel time and energy consumption            | Approximate dynamic programming based algorithm            |
| Yang et al. (2017) | Passenger flow control and bus-bridging                              | –                   | No  | Passenger waiting time                                               | CPLEX                                                      |
| Wang et al. (2018) | Train schedule and rolling stock circulation                         | Cycle               | No  | Headway variation, load factor variation, number of depot operations | CPLEX and iterative approach                               |
| Shi et al. (2018)  | Train schedule and passenger flow control                            | Cycle               | No  | Passenger waiting time in station halls and platforms                | CPLEX and heuristic based on local search algorithm        |
| Mo et al. (2019b)  | Train schedule                                                       | Semicycle           | Yes | Passenger waiting time and energy cost                               | Modified tabu search                                       |
| Liu et al. (2020)  | Train schedule, passenger flow control and rolling stock circulation | Cycle               | No  | Total number of waiting passengers and train service connection      | CPLEX and Lagrangian relaxation based algorithm            |
| Yue et al. (2017)  | Train frequency, train schedule and rolling stock circulation        | Cycle and semicycle | No  | Passenger waiting time, operating costs and infeasible services      | CPLEX and heuristic based on simulated annealing algorithm |
| This study         | Train schedule, passenger flow control and rolling stock circulation | Semicycle           | Yes | Passenger waiting time and operating costs                           | CPLEX and heuristic based on tabu search algorithm         |

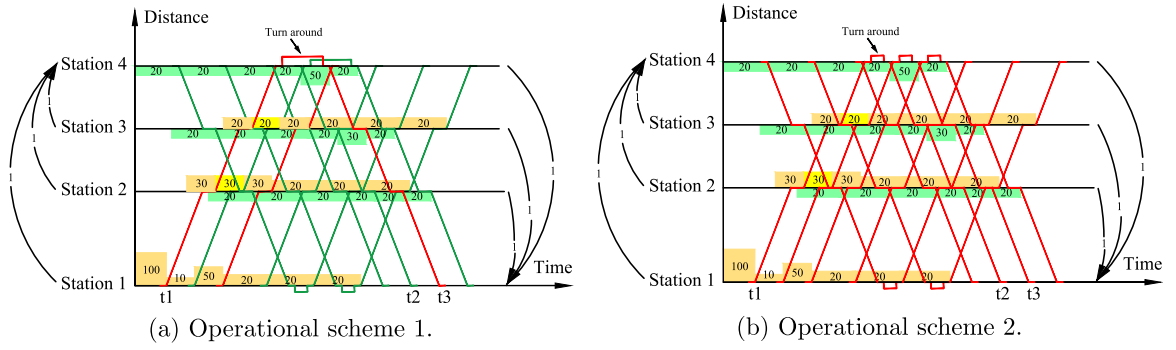


**Fig. 1.** The structure of the considered metro corridor in this study.

two alternative operations, i.e., (1) turning around to undertake a downstream service or (2) returning to depot 2 for other service assignment. Then, the semicircle operation of the train terminates. Next, we define two periods in the operation process, i.e., online period and offline period. In detail, the online period is defined as the operation from one terminal station to another (i.e., from stations 1 to  $|S|$  or from stations  $|S| + 1$  to  $2|S|$ ), which corresponds to the process of executing services. The offline period includes two types of operations, i.e., running between the depot and its connected end station, and turning around at stations  $|S|$  and  $2|S|$ . Thus, a “semicircle” operation consists of an online period and one/two offline periods. In this sense, two semicircle operations usually form a circle operation. This type of operation is commonly adopted in recent studies (e.g., Gao et al., 2016; Mo et al., 2019a).

In this metro corridor, we assume that at least one side is oversaturated during the peak hours, and the passenger arrival ratio is time-dependent. The considered time horizon includes at least one peak period and partial of the subsequent off-peak period. The aim of this study is to optimize the operation strategy from the perspective of system optimization, in which four components, including train schedule, rolling stock circulation plan, passenger flow control and multiple types of rolling stocks, are taken into account to match the passenger flow and train operations. This process is fulfilled by introducing a series of coupling constraints between different components. Note that since this study introduces several types of rolling stocks, we next present an illustration to show the benefits brought by reasonably utilizing different types of rolling stocks.

Here, we consider a bidirectional metro corridor with 4 stations and 3 sections in each direction, where the operating direction from station 1 to station 4 is defined as the upstream direction and that from station 4 to station 1 is the downstream direction. A total of 6 services are assumed to run in each direction. To demonstrate the effectiveness and necessity of considering different types of rolling stocks, two operation schemes are considered. Fig. 2(a) shows the operational scheme with the consideration of two types of rolling stocks, and Fig. 2(b) shows the situation in which all services are implemented by high-capacity rolling stocks (commonly used by many metro systems). In Fig. 2, red and green lines denote large-capacity and low-capacity rolling stocks, respectively, with loading capacities of 100 and 60. The fixed costs for using these trains are 100 and 60, respectively. The bars filled with orange and



**Fig. 2.** An illustration of different operational schemes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

green colors denote passengers on upstream platforms and downstream platforms, respectively. In particular, the bars filled with yellow indicate the passengers waiting for the second time on the platform (i.e., these passengers need to wait for the next service due to the capacity limitations of the present arriving service). The origin and destination (OD) of passengers are set as following: (1,4),(2,4),(3,4),(4,1),(3,1),(2,1). Here, (1,4) represents that all passengers at station 1 are destined for station 4. With the current passenger demand, it is easy to see that the two schemes can transport the same number of passengers. However, the operating costs are quite different due to the consideration of operating different types of rolling stocks. In Fig. 2(a), the corresponding fixed cost of services is  $3 \cdot 100 + 9 \cdot 60 = 840$ , which is increased to  $12 \cdot 100 = 1200$  for the scheme in Fig. 2(b). In comparison to only operating high-capacity rolling stocks, the operating cost for the plan with multiple types of rolling stocks can be reduced by 30%. However, compared with using one type of rolling stocks, operating different types of rolling stocks typically increases the difficulties of operations since except for the connection time, the types of rolling stocks also should be considered in the process of rolling stock circulation. For instance, service  $t_1$  must connect to service  $t_3$  instead of service  $t_2$  since  $t_1$  and  $t_2$  are different types of rolling stocks in Fig. 2(a). While, in Fig. 2(b), service  $t_1$  can be easily connected to service  $t_2$  as the types of rolling stocks for these two services are the same. Mathematically, we need to introduce additional decision variable as an indicator to represent which type of rolling stock is selected for each service and create a detailed connection plan for different services, which unavoidably increases the decision complexity of the problem.

Moreover, considering the common situation that a large number of passengers accumulate at stations during peak hours due to the tremendous increase of passenger demand, it is necessary to adopt passenger flow control strategy to reduce potential risks in these situations. Thus, this study considers a joint optimization approach to simultaneously decide the specific departure and arrival times of each train at each station, the type of rolling stock for each service, the circulation plan of different rolling stocks, and the passenger flow control strategy at each station. Next, some assumptions are presented for the convenience of formulating the problem.

**Assumption 1.** The boarding behavior of passengers is simplified, i.e., the walking time from the station hall to platform is not taken into account. The passengers are required to wait at the station hall until the service arrives, and then they directly board the arrival service according to the passenger flow control strategy.

**Assumption 2.** The section running time and station dwell time are not treated as decision variables, which are input parameters predetermined for each service. This assumption also coincides with practical operations, as the scheduling process is usually related to optimizing the headway of adjacent services.

**Assumption 3.** For modeling convenience, the turn-around capacity of each end station is assumed to be large enough. With this assumption, each train is allowed to turn around at any time, if necessary.

### 3. Mathematical model

This section aims to formulate the considered problem as a rigorous optimization model. The following discussion focuses on introducing each component of the model, including parameters, decision variables, systematic constraints, objective functions and complexity of the formulation. For modeling convenience, Table 2 lists all the relevant subscripts and parameters used in the formulations.

In the joint optimization model, we aim to specify the train schedule, passenger flow control strategy and rolling stock circulation plan. Thus, seven classes of decision variables are considered in this study.

- $t_{l,k}^a$ : arrival time of service  $l$  at station  $k$ .
- $t_{l,k}^d$ : departure time of service  $l$  at station  $k$ .

**Table 2**  
Subscripts and parameters used in the formulation.

| Notations                      | Definition                                                                                                                                                                                   |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\bar{N}$                      | set of services in the upstream direction, $\bar{N} = \{1, 2, \dots,  \bar{N} \}$ .                                                                                                          |
| $\underline{N}$                | set of services in the downstream direction, $\underline{N} = \{1, 2, \dots,  \underline{N} \}$ .                                                                                            |
| $l, l'$                        | index of services, $l, l' \in \bar{N} \cup \underline{N}$ .                                                                                                                                  |
| $\bar{S}$                      | set of involved metro stations in the upstream direction, $\bar{S} = \{1, 2, \dots,  \bar{S} \}$ .                                                                                           |
| $\underline{S}$                | set of involved metro stations in the downstream direction, $\underline{S} = \{ \bar{S}  + 1,  \bar{S}  + 2, \dots, 2 \bar{S} \}$ .                                                          |
| $k, v, s$                      | index of stations, $k, v, s \in \bar{S} \cup \underline{S}$ ; these indexes can also represent the sections (e.g., section $k$ represents the section from station $k$ to station $k + 1$ ). |
| $J$                            | set of rolling stock types, $J = \{1, 2, \dots,  J \}$ .                                                                                                                                     |
| $j$                            | index of rolling stock types, $j \in J$ .                                                                                                                                                    |
| $T$                            | set of timestamps in the discretized time horizon, $T = \{1, 2, \dots, q\}$ .                                                                                                                |
| $t, \tau$                      | index of timestamps, $t, \tau \in T$ .                                                                                                                                                       |
| $t_{unit}$                     | time length between two adjacent timestamps.                                                                                                                                                 |
| $t_{l,k}^r$                    | running time for service $l$ on section $k$ .                                                                                                                                                |
| $t_{l,k}^w$                    | dwell time for service $l$ at station $k$ .                                                                                                                                                  |
| $t_{l,k}^{(f)}$                | the times when the first service arrives at the origin station in the upstream and downstream directions.                                                                                    |
| $t_{l,k}^{(a)}$                | the earliest times that the last service arrives at the origin station in the upstream and downstream directions.                                                                            |
| $h_{max}$                      | the maximum headway for two consecutive services.                                                                                                                                            |
| $h_{min}$                      | the minimum headway for two consecutive services.                                                                                                                                            |
| $t_{l,k}^{(d)}$                | the running time from depot 1 and depot 2 to their connected stations, respectively.                                                                                                         |
| $t_{l,k}^{(e)}$                | the running time from end stations to depot 2 and depot 1, respectively.                                                                                                                     |
| $t_{l,k}^{(roll)}$             | the running time for rolling stocks to turn around in the upstream and downstream directions, respectively.                                                                                  |
| $t_{l,k}^{(roll)}$             | the maximum time for the rolling stock to complete the turn-around operations.                                                                                                               |
| $\eta_k(t)$                    | the number of arrival passengers at station $k$ during timestamp $t$ .                                                                                                                       |
| $\delta_{v,k}(t)$              | the ratio of passengers with destination station $k$ that board at station $v$ over time $t$ .                                                                                               |
| $C_j^{max}$                    | the maximum capacity of a $j$ -type rolling stock.                                                                                                                                           |
| $\bar{CN}_j, \underline{CN}_j$ | the numbers of $j$ -type rolling stocks at depot 1 and depot2, respectively.                                                                                                                 |
| $M_j^{cost}$                   | the fixed cost of executing a service by $j$ -type rolling stock.                                                                                                                            |
| $M^{out}$                      | the movement cost of a rolling stock exiting from the depots.                                                                                                                                |
| $M$                            | a large positive number.                                                                                                                                                                     |

$y_{l,j}$ : indicator of service  $l$  undertaken by  $j$ -type of rolling stock, i.e.,  $= 1$  if service  $l$  is executed by  $j$ -type of rolling stock;  $= 0$ , otherwise.

$P_k^{in}(t)$ : number of passengers allowed to enter the platform at station  $k$  over timestamp  $t$ .

$\alpha_l$ : indicator of whether the dedicated rolling stock for service  $l$  comes from the depot, i.e.,  $= 1$  if the rolling stock for service  $l$  does not come from the depot;  $= 0$ , otherwise.

$\beta_l$ : indicator showing whether the dedicated rolling stock for service  $l$  returns to the depot,  $= 1$  if the rolling stock for executing service  $l$  does not return to the depot;  $= 0$ , otherwise.

$\gamma_{l,l'}$ : indicator of the connection process,  $= 1$  if the rolling stock executing service  $l$  can turn around and continue to execute service  $l'$ ;  $= 0$ , otherwise.

In addition, two classes of extra auxiliary variables are introduced to formulate the model into a tractable form. The first class is associated with train scheduling auxiliary variables, which indicate the passenger boarding activities and the arrival time of each service at each station. The second class refers to auxiliary variables for passenger flow control strategy, including the numbers of waiting, alighting and boarding passengers. The detailed auxiliary variables are depicted as follows.

$b_{l,k}(t)$ : indicator of boarding activities,  $= 1$  if passengers at station  $k$  can board service  $l$  in time interval  $[t, t + 1)$ ;  $= 0$ , otherwise.

$z_{l,k}(t)$ : indicator of service arrival,  $= 1$  if service  $l$  arrives at station  $k$  at or before timestamp  $t$ ;  $= 0$ , otherwise.

$P_k^{wa}(t)$ : the number of waiting passengers in the station hall at station  $k$  at timestamp  $t$ .

$P_{l,k}^{al}$ : the number of alighting passengers from service  $l$  at station  $k$ .

$P_{l,k}^{on}$ : the number of loading passengers on service  $l$  between stations  $k$  and  $k + 1$ .

$P_{l,k}^{bo}$ : the total number of boarding passengers for service  $l$  at station  $k$ .

$P_{l,k,s}^{bo}$ : the number of boarding passengers for service  $l$  traveling from stations  $k$  to  $s$ .

### 3.1. Systematic constraints

Next, a series of constraints are formulated to satisfy the passenger demand with different types of rolling stocks and guarantee the feasibility and safety of the train schedule and turn-around operations. As this study investigates a joint optimization problem, the constraints are formulated in line with different components, and the coupling constraints between different parts are also established.



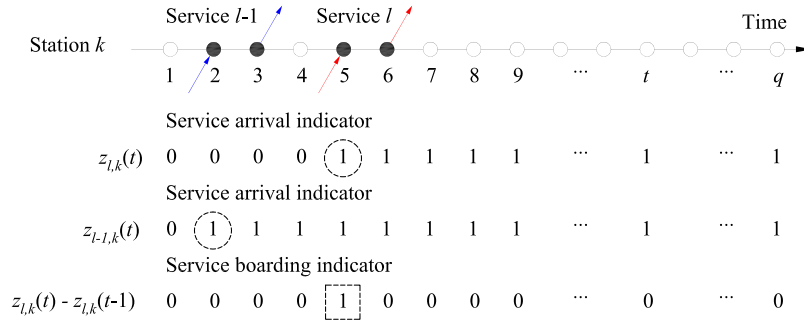


Fig. 3. An illustration of timetabling variables.

### 3.1.1. Train schedule constraints

In practice, the schedule, which provides instructions for all the involved services, is the foundation for train operations of a metro system. We first formulate constraints for train scheduling, which effectively restricts the arrival and departure times of each service at each station.

#### (i) Arrival and departure time constraints

This study assumes that the section running time and station dwell time are pre-given. To keep this condition, the arrival and departure constraints for the involved services can be formulated as follows.

$$t_{l,k}^a = t_{l,k-1}^d + t_{l,k-1}^r, \forall l \in \bar{N}, k, k-1 \in \bar{S} \text{ or } \forall l \in \underline{N}, k, k-1 \in \underline{S}. \quad (1)$$

$$t_{l,k}^d = t_{l,k}^a + t_{l,k}^w, \forall l \in \bar{N}, k \in \bar{S} \text{ or } \forall l \in \underline{N}, k \in \underline{S}. \quad (2)$$

In Eq. (1), the arrival time of service  $l$  at station  $k$  is equal to the sum of the arrival time of service  $l$  at station  $k-1$  and the travel time of this service on section  $k-1$ . Similarly, the arrival time of service  $l$  at station  $k$  plus its dwell time at station  $k$  should be the departure time of service  $l$  at station  $k$ .

In addition, considering the practical requirements of services, the following constraints should be imposed to limit the arrival times of the first and last services at their origin stations in the upstream direction.

$$t_{1,1}^a = t_{1,1}^{(fi)}, \quad t_{1,1}^a \geq t_{1,1}^{(la)}. \quad (3)$$

Similarly, the arrival time of the first and last services at their origin stations in the downstream direction should satisfy the following constraints.

$$t_{1,|S|+1}^a = t_{1,|S|+1}^{(fi)}, \quad t_{1,|S|+1}^a \geq t_{1,|S|+1}^{(la)}. \quad (4)$$

Note that the train schedule is also the foundation for specifying dynamics of passengers since the passengers are only allowed to board and alight the service dwelling at the station. To combine the train schedule and the boarding/alighting passengers, this study introduces an auxiliary binary variable  $z_{l,k}(t)$  to represent the arrival activities of each service, i.e.,  $z_{l,k}(t) = 1$  indicates that service  $l$  arrives at station  $k$  at or before timestamp  $t$ , and  $= 0$  otherwise. This representation method is also adopted in some existing studies (e.g., Niu et al., 2015; Mo et al., 2019b). Take Fig. 3 as an illustration. Service  $l$  arrives at station  $k$  at timestamp 5 and departs from station  $k$  at timestamp 6; therefore, we have  $z_{l,k}(5) = 1$ . In this presentation, we can easily set up the relationship between this variable  $z_{l,k}(t)$  and the arrival time  $t_{l,k}^a$ . That is,  $z_{l,k}(t) - z_{l,k}(t-1) = 1$  always holds on timestamp  $t_{l,k}^a$ , leading to the following constraint.

$$t_{l,k}^a = \sum_{t \in T} t \cdot [z_{l,k}(t) - z_{l,k}(t-1)], \forall l \in \bar{N}, k \in \bar{S}, t, t-1 \in T \text{ or } \forall l \in \underline{N}, k \in \underline{S}, t, t-1 \in T. \quad (5)$$

In this sense,  $z_{l,k}(t) - z_{l,k}(t-1) = 1$  can be regarded as the indicator of service  $l$  dwelling at station  $k$ . As the passenger at a station can only board a service that is dwelling at that station, we can use this indicator as the boarding time indicator according to Assumption 1. In particular, for timestamp 1, this indicator should be expressed as  $z_{l,k}(1)$ . Here, we adopt  $b_{l,k}(t)$  to denote the service boarding indicator, which can be formulated as follows.

$$b_{l,k}(t) = \begin{cases} z_{l,k}(t), & \text{if } t = 1 \\ z_{l,k}(t) - z_{l,k}(t-1), & \text{otherwise} \end{cases}$$

$$\forall l \in \bar{N}, k \in \bar{S}, t \in T \text{ or } \forall l \in \underline{N}, k \in \underline{S}, t \in T. \quad (6)$$

In addition, since this arrival indicator variable is nondecreasing, it should satisfy the following constraint.

$$z_{l,k}(t-1) \leq z_{l,k}(t), \forall l \in \bar{N}, k \in \bar{S}, t, t-1 \in T \text{ or } \forall l \in \underline{N}, k \in \underline{S}, t, t-1 \in T. \quad (7)$$

## (ii) Headway constraints

To guarantee safe operations of the metro system, the headway between two adjacent services should be limited within an allowable range. As the dwell time at each station and running time on each section are pre-specified constants, we can restrict the arrival times of adjacent services at the origin station to guarantee safe headway in the upstream and downstream directions, respectively, which can be formulated as the following constraints.

$$h_{\min} \leq t_{l,1}^a - t_{l-1,1}^a \leq h_{\max}, \quad \forall l, l-1 \in \bar{N}, \quad (8)$$

$$h_{\min} \leq t_{l,|S|+1}^a - t_{l-1,|S|+1}^a \leq h_{\max}, \quad \forall l, l-1 \in \underline{N}, \quad (9)$$

where  $h_{\min}$  and  $h_{\max}$  are the lower and upper bounds for the headway constraints to guarantee the safety and efficiency of the metro operations.

## 3.1.2. Constraints with respect to passenger flow control strategy

This study considers an oversaturated situation in the metro line, where at least one peak period is involved. We also aim to optimize the passenger flow control strategy to collaboratively reduce the passenger waiting time among different stations in order to guarantee operation safety. Some system constraints are formulated below to guarantee the feasibility of the control strategy, including the passenger waiting and passenger loading constraints.

**(i) Passenger waiting constraints.** At each station, the arrival passengers are required to wait in the station hall for the arrival of services. Then, we need to calculate the volume of passengers waiting at each station to capture the dynamics of passengers' boarding and alighting activities. Typically, the total number of waiting passengers at station  $k$  and timestamp  $t$  is equal to the number of all arrival passengers minus the number of passengers entering the platform before timestamp  $t$ , which can be expressed as follows.

$$P_k^{wa}(t) = \sum_{\tau=1}^t \eta_k(\tau) - \sum_{\tau=1}^t P_k^{in}(\tau), \quad \forall k \in \bar{S}, t \in T \text{ or } \forall k \in \underline{S}, t \in T. \quad (10)$$

In addition, the number of waiting passengers at any station and timestamp should not be negative, leading to the following constraints.

$$P_k^{wa}(t) \geq 0, \quad \forall k \in \bar{S}, t \in T \text{ or } \forall k \in \underline{S}, t \in T. \quad (11)$$

Finally, all the passengers at each station should be served, leading to the following constraints.

$$\sum_{t \in T} \eta_k(t) = \sum_{t \in T} P_k^{in}(t), \quad \forall k \in \bar{S} \text{ or } \forall k \in \underline{S}. \quad (12)$$

**(ii) Passenger loading constraints.**

According to [Assumption 1](#), if a service arrives at a station, passengers will enter the platform and board this service according to the passenger flow control strategy. With the auxiliary variable  $b_{l,k}(t)$ , this progress can be characterized as follows.

$$P_{l,k,s}^{bo} = \sum_{t \in T} b_{l,k}(t) \cdot P_k^{in}(t) \cdot \delta_{k,s}(t), \quad \forall l \in \bar{N}, k < s \in \bar{S} \text{ or } \forall l \in \underline{N}, k < s \in \underline{S}. \quad (13)$$

Here,  $\delta_{k,s}(t)$  is a time-dependent coefficient that indicates the ratio of passengers with destination station  $s$ . Additionally, since both  $b_{l,k}(t)$  and  $P_k^{in}(t)$  are decision variables, Eq. (13) is a nonlinear constraint. For solution convenience, this constraint is linearized in Section 3.3.

Next, we calculate the total number of passengers who board service  $l$  at station  $k$ , which corresponds to the total number of passengers with a trip from station  $k$  to station  $s$ , as shown below.

$$P_{l,k}^{bo} = \sum_{s>k} P_{l,k,s}^{bo}, \quad \forall l \in \bar{N}, k < s \in \bar{S} \text{ or } \forall l \in \underline{N}, k < s \in \underline{S}. \quad (14)$$

Likewise, the total number of alighting passengers at station  $k$  from service  $l$  is the sum of the passengers in service  $l$  with destination station  $k$ , formulated as follows.

$$P_{l,k}^{al} = \sum_{v<k} P_{l,v,k}^{bo}, \quad \forall l \in \bar{N}, v < k \in \bar{S} \text{ or } \forall l \in \underline{N}, v < k \in \underline{S}. \quad (15)$$

Here, we explicitly characterize the dynamic evolution of passengers in services due to the passengers' boarding and alighting activities. Since there is no passenger boarding or alighting activity while the service runs on the section, the number of in-vehicle passengers for service  $l$  on section  $k$  (i.e.,  $P_{l,k}^{on}$ ) can be expressed as the number of loaded passengers when service  $l$  departs from station  $k$ . On one hand, since there is no passenger alighting from the service, the number of loaded passengers in service  $l$  at the original station is actually the number of boarding passengers. On the other hand, the number of loaded passengers on service  $l$  when it departs from other station  $k$  is equal to the number of loaded passengers when departing from the previous station  $k-1$  plus the number of boarding passengers at station  $k$  and minus the number of alighting passengers at station  $k$ .

$$P_{l,k}^{on} = \begin{cases} P_{l,k}^{bo}, & \text{if } k = 1 \\ P_{l,k-1}^{on} + P_{l,k}^{bo} - P_{l,k}^{al}, & \text{otherwise} \end{cases}$$



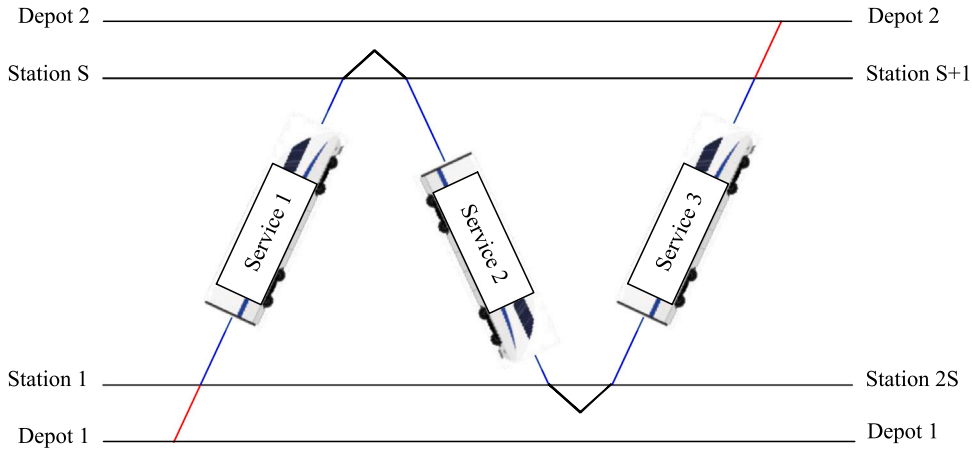


Fig. 4. An illustration of auxiliary variables for rolling stock circulation.

$$\forall l \in \bar{N}, k \in \bar{S} \text{ or } \forall l \in \underline{N}, k \in \underline{S}. \quad (16)$$

Typically, the number of boarding passengers is limited by the remaining loading capacity of the service. Since this study provides multiple types of rolling stocks, the maximum loading capacity of each service is dependent on the type of rolling stock. Specifically, at the origin station, the available loading capacity equals to the maximum capacity of the rolling stock. At other stations, the remaining loading capacity is related to the maximum capacity, the number of loaded passengers at the previous station and the number of alighting passengers at the current station.

$$P_{l,k}^{bo} \leq \begin{cases} \sum_{j \in J} y_{l,j} \cdot C_j^{max}, & \text{if } k = 1 \\ \sum_{j \in J} y_{l,j} \cdot C_j^{max} - P_{l,k-1}^{on} + P_{l,k}^{al}, & \text{otherwise} \end{cases}$$

$$\forall l \in \bar{N}, k \in \bar{S} \text{ or } \forall l \in \underline{N}, k \in \underline{S}. \quad (17)$$

In addition, since all the passengers should be served by the metro system, the following constraints should be formulated.

$$\sum_{i \in T} P_k^{in}(t) = \sum_{l \in \bar{N}} P_{l,k}^{bo}, \quad \forall k \in \bar{S}. \quad (18)$$

$$\sum_{i \in T} P_k^{in}(t) = \sum_{l \in \underline{N}} P_{l,k}^{bo}, \quad \forall k \in \underline{S}. \quad (19)$$

What's more, each service can only be executed by one type of rolling stock (i.e., the train capacity is fixed during its operation), leading to the following constraints.

$$\sum_{j \in J} y_{l,j} = 1, \quad \forall l \in \underline{N} \text{ or } l \in \bar{N}. \quad (20)$$

### 3.1.3. Constraints related to rolling stock circulation

This study considers multiple types of rolling stocks and two depots at both end of the line for rolling stock circulation planning. This is one of the significant focuses to deal with the large tidal passenger demand. Next, several constraints related to the rolling stock circulation plan are developed to guarantee the successful connection for each type of rolling stocks.

**(i) Auxiliary variable constraints.** For modeling convenience, we introduce a set of variables  $\alpha_l$ ,  $\beta_l$  and  $\gamma_{l,l'}$  to capture the off-line operations of rolling stocks, including departing from the depot, returning to the depot and turnaround for the subsequent operations. In detail,  $\alpha_l = 1$  shows that the rolling stock executing service  $l$  is not from the depot, i.e., service  $l$  is executed by a previous online rolling stock, and  $\alpha_l = 0$  otherwise.  $\beta_l = 1$  indicates that the rolling stock will not return to the depot after executing service  $l$ , and  $\beta_l = 0$  otherwise.  $\gamma_{l,l'} = 1$  implies that the rolling stock will not return to the depot after executing service  $l$ , as it will undertake service  $l'$  in the opposite direction.

For clarity, we next take Fig. 4 as an example to illustrate it. A rolling stock leaves from depot 1 and then executes service 1, i.e.,  $\alpha_1 = 0$ . Upon arriving at terminal station  $S$ , this rolling stock turns around to execute service 2, which corresponds to  $\beta_1 = 1$ ,  $\alpha_2 = 1$  and  $\gamma_{1,2} = 1$ . Afterwards, the connection of services 2 and 3 leads to  $\beta_2 = 1$ ,  $\alpha_3 = 1$  and  $\gamma_{2,3} = 1$ . Finally, this rolling stock returns to depot 2, i.e.,  $\beta_3 = 0$ . The relationship among  $\alpha_l$ ,  $\beta_l$  and  $\gamma_{l,l'}$  can be formulated as follows.

Firstly, for a service  $l'$  in the upstream direction, at most one rolling stock can be scheduled to continue executing it after finishing the previous services in the downstream direction. Nevertheless, if no rolling stock that finishes previous services can be used to

execute service  $l'$ , a dedicated rolling stock should depart from depot 1 to execute this service. The following constraints can be formulated to guarantee these operations.

$$0 \leq \sum_{l' \in \bar{N}} \gamma_{l,l'} \leq 1, \forall l' \in \bar{N}. \quad (21)$$

$$\alpha_{l'} = \sum_{l \in \underline{N}} \gamma_{l,l'}, \forall l' \in \bar{N}. \quad (22)$$

Likewise, similar constraints also hold for the rolling stock to execute a service  $l$  in the downstream direction. That is,

$$0 \leq \sum_{l' \in \bar{N}} \gamma_{l',l} \leq 1, \forall l \in \underline{N}. \quad (23)$$

$$\alpha_l = \sum_{l' \in \bar{N}} \gamma_{l',l}, \forall l \in \underline{N}. \quad (24)$$

Secondly, when a rolling stock completes a service  $l'$  in the upstream direction, at most one subsequent service can be successively operated by this rolling stock. When this rolling stock is not assigned to execute a subsequent service, it will return to depot 1. This process can be characterized by the following constraints.

$$0 \leq \sum_{l \in \underline{N}} \gamma_{l',l} \leq 1, \forall l' \in \bar{N}. \quad (25)$$

$$\beta_{l'} = \sum_{l \in \underline{N}} \gamma_{l',l}, \forall l' \in \bar{N}. \quad (26)$$

With respect to the rolling stock finishing a service  $l$  in the downstream direction, similar constraints also hold, as shown below.

$$0 \leq \sum_{l' \in \bar{N}} \gamma_{l,l'} \leq 1, \forall l \in \underline{N}. \quad (27)$$

$$\beta_l = \sum_{l' \in \bar{N}} \gamma_{l,l'}, \forall l \in \underline{N}. \quad (28)$$

**(ii) Rolling stock type connection constraints.** Since multiple types of rolling stocks are considered in this study to match the passenger flow and the provided services, in the planning process, if a rolling stock is scheduled to execute service  $l'$  when it finishes service  $l$ , these two services should be connected by the same type of rolling stocks. To guarantee the feasibility of service connections, we define the relationships among variables  $y_{l,j}$ ,  $y_{l',j}$  and  $\gamma_{l,l'}$  by the following constraints.

$$\begin{cases} y_{l,j} - y_{l',j} \leq M \cdot (1 - \gamma_{l,l'}), \\ y_{l,j} - y_{l',j} \geq -M \cdot (1 - \gamma_{l,l'}), \end{cases}$$

$$\forall l \in \underline{N}, l' \in \bar{N}, j \in J \text{ or } \forall l \in \bar{N}, l' \in \underline{N}, j \in J. \quad (29)$$

In this set of constraints,  $\gamma_{l,l'} = 1$  indicates that the dedicated rolling stock connects services  $l$  and  $l'$  successfully, leading to the same type of rolling stocks for services  $l$  and  $l'$ , i.e.,  $y_{l,j} = y_{l',j}$ ; otherwise, no restrictions are imposed on the values of decision variables  $y_{l,j}$  and  $y_{l',j}$ .

**(iii) Turn-around constraints.** When a rolling stock is scheduled to execute a following service after it finishing a service in the opposite direction, some time is required for this rolling stock to complete the turn-around operation at the designated terminal station. The following constraint holds for the turn-around operation from station  $2|S|$  to station 1.

$$t_{l',1}^a - t_{l,2|S|}^d \geq t_{\text{roll}}^{(roll)} - M \cdot (1 - \gamma_{l,l'}), \forall l \in \underline{N}, l' \in \bar{N}. \quad (30)$$

$$t_{l',1}^a - t_{l,2|S|}^d \leq t_{\text{max}}^{(roll)} + M \cdot (1 - \gamma_{l,l'}), \forall l \in \underline{N}, l' \in \bar{N}. \quad (31)$$

That is, the time duration between a rolling stock's arrival at station 1 (for executing service  $l'$  in the upstream direction) and its departure from station  $2|S|$  (when it finishes service  $l$  in the downstream direction) should be constrained in a pre-specified time range  $[t_{\text{roll}}^{(roll)}, t_{\text{max}}^{(roll)}]$ .

Similarly, the turn-around operation from station  $|S|$  to station  $|S|+1$  for the rolling stock to execute service  $l'$  in the downstream direction after it finishing service  $l$  in the upstream direction can be guaranteed by the following constraints.

$$t_{l,|S|+1}^a - t_{l',|S|}^d \geq t_{\text{roll}}^{(roll)} - M \cdot (1 - \gamma_{l,l'}), \forall l \in \underline{N}, l' \in \bar{N}. \quad (32)$$

$$t_{l,|S|+1}^a - t_{l',|S|}^d \leq t_{\text{max}}^{(roll)} + M \cdot (1 - \gamma_{l,l'}), \forall l \in \underline{N}, l' \in \bar{N}. \quad (33)$$

**(iv) Available rolling stock constraints.** Generally, the number of required rolling stocks is usually much less than the number of online services since each rolling stock usually can be scheduled to undertake a set of services. The number of available rolling stocks (of various types) at different depots should be enough for practical operations according to the designed schedule, i.e., the

number of on-line rolling stocks cannot exceed the number of available rolling stocks. Practically, the number of each type of rolling stocks at each depot varies along with the dynamic departure and arrival operations of rolling stocks. For instance, if a rolling stock is scheduled to undertake a service, the number of inventory rolling stocks in the corresponding depot will decrease when this rolling stock departs from the depot. Additionally, the offline rolling stocks scheduled to return to the depot will increase the number of inventory rolling stocks. Typically, in this process, the number of available rolling stocks at each depot should not be negative in order to maintain the regular operations of the metro system. For this purpose, the number of different types of rolling stocks in depot 1 can be restricted by the following constraints, with the aforementioned auxiliary variables  $z_{l,k}(t)$ ,  $\beta_l$  and  $\alpha_{l'}$ .

$$\overline{CN}_j + \sum_{l \in \underline{N}} (1 - \beta_l) \cdot y_{l,j} \cdot z_{l,2|S|}(t - t_{l,2|S|}^w - \underline{t}^{(re)}) - \sum_{l' \in \underline{N}} (1 - \alpha_{l'}) \cdot y_{l',j} \cdot z_{l',1}(t + \underline{t}^{(le)}) \geq 0, \\ \forall j \in J, t, t - t_{l,2|S|}^w - \underline{t}^{(re)}, t + \underline{t}^{(le)} \in T. \quad (34)$$

In Eq. (34),  $\overline{CN}_j$  denotes the initial number of inventory  $j$ -type rolling stocks in depot 1:  $1 - \beta_l = 1$  if a rolling stock returns to depot 1 after finishing service  $l$ , and  $1 - \beta_l = 0$  otherwise. Moreover,  $1 - \alpha_{l'} = 1$  if a rolling stock leaves from depot 1 to implement service  $l'$ , and  $1 - \alpha_{l'} = 0$  otherwise. Additionally,  $\underline{t}^{(re)}$  denotes the running time from the terminal station  $2|S|$  to depot 1, and  $\underline{t}^{(le)}$  denotes the running time from depot 1 to origin station 1. Since we use  $z_{l,k}(t)$  to represent the arrival indicator of service  $l$  at station  $k$ , the dwell time at station  $2|S|$  should be subtracted. So, the second expression  $\sum_{l \in \underline{N}} (1 - \beta_l) \cdot y_{l,j} \cdot z_{l,2|S|}(t - t_{l,2|S|}^w - \underline{t}^{(re)})$  and third term  $\sum_{l' \in \underline{N}} (1 - \alpha_{l'}) \cdot y_{l',j} \cdot z_{l',1}(t + \underline{t}^{(le)})$  respectively represent the number of arrival and departure rolling stocks at time  $t$ . Likewise, the inventory rolling stock constraints in depot 2 can be formulated as follows.

$$\underline{CN}_j + \sum_{l' \in \underline{N}} (1 - \beta_{l'}) \cdot y_{l',j} \cdot z_{l',|S|}(t - t_{l',|S|}^w - \underline{t}^{(re)}) - \sum_{l \in \underline{N}} (1 - \alpha_l) \cdot y_{l,j} \cdot z_{l,|S|+1}(t + \underline{t}^{(le)}) \geq 0, \\ \forall j \in J, t, t - t_{l',|S|}^w - \underline{t}^{(re)}, t + \underline{t}^{(le)} \in T. \quad (35)$$

### 3.2. Objective function

In reality, the evaluation of operation plans is closely related to the satisfaction of passengers and the costs of metro companies. On the one hand, to provide better services for passengers, it is desirable to maximize the service quality characterized by the minimum passenger waiting time. On the other hand, the operating costs should be minimized for the generated train schedule and rolling stock circulation plan to improve the operation efficiency. With these considerations, the objective function takes three parts into account, namely, passenger waiting time in station halls (denoted by  $WT$ ), fixed costs of online services (denoted by  $CR$ ) and movement costs of trains exiting from the depot (denoted by  $CS$ ), to evaluate the generated operation plans. The following discussion details each part explicitly.

Recall that we discretize the considered time horizon into a series of timestamps for modeling convenience. Then, the passenger waiting time in station halls can be calculated by the sum of waiting passengers at each station over all timestamps, formulated as below.

$$WT = \sum_{t \in T} \sum_{k \in S \cup \underline{S}} P_k^{wa}(t) \cdot t_{unit}. \quad (36)$$

In addition, as multiple types of rolling stocks are considered, different types of rolling stocks correspond to diverse fixed costs for executing services. In the modeling process, we introduce a binary decision variable  $y_{l,j}$  to indicate whether service  $l$  chooses a  $j$ -type rolling stock. Thus, in the formulation, the product of  $y_{l,j}$  and  $M_j^{cost}$  denotes the fixed costs of using a  $j$ -type rolling stock to execute service  $l$ . In this sense, the total fixed costs of services can be calculated as follows.

$$CR = \sum_{j \in J} \sum_{l \in \underline{N} \cup \underline{N}} y_{l,j} \cdot M_j^{cost}. \quad (37)$$

Note that, as the number of services is fixed, minimizing the number of rolling stocks exiting from the depot means that more services can be executed by the same rolling stock, which affects the total costs of the rolling stock circulation plan. Thus, we finally focus on the movement of rolling stocks that exit from the depots, listed below.

$$CS = \sum_{l \in \underline{N} \cup \underline{N}} [(1 - \alpha_l) \cdot M^{out}]. \quad (38)$$

Moreover, the fixed costs of online services and movement costs of rolling stocks exiting from the depot can be merged into the operating costs of metro companies. Then, by introducing two price coefficients  $\lambda_1$  and  $\lambda_2$  for the passenger waiting time and operating costs, the multiple objectives can be regrouped into a single-objective function.

$$\min F = \lambda_1 \cdot \sum_{t \in T} \sum_{k \in S \cup \underline{S}} P_k^{wa}(t) \cdot t_{unit} + \lambda_2 \cdot \left( \sum_{j \in J} \sum_{l \in \underline{N} \cup \underline{N}} y_{l,j} \cdot M_j^{cost} + \sum_{l \in \underline{N} \cup \underline{N}} [(1 - \alpha_l) \cdot M^{out}] \right) \quad (39)$$

In practice, the values of these two price coefficients are dominated by the preferences of decision makers and realistic operation situations. For instance, if decision makers pay more attention to passenger satisfaction, coefficient  $\lambda_1$  can be larger than  $\lambda_2$ ; otherwise, a larger value of  $\lambda_2$  should be adopted. Additionally, this study also conducts a series of numerical experiments to find a suitable combination of these two coefficients to balance the interests of metro companies and passengers (see Section 5.1 for more details).

### 3.3. Linearization of nonlinear formulations

In the proposed joint optimization model, Eqs. (13), (34) and (35) have nonlinear forms, making it difficult to obtain the solution via some exact methods (e.g., branch-and-bound, branch-and-price) or optimization solvers (e.g., CPLEX, GUROBI). With the help of linearization methods proposed by Bemporad and Morari (1999) in Appendix, we convert these equations into more tractable linear forms.

**Lemma 3.1.** In Eq. (13), as  $\delta_{k,s}(t)$  is a pre-given constant parameter,  $\delta_{k,s}(t) \cdot P_k^{in}(t)$  is a continuous variable in the model. By introducing an auxiliary variable  $\omega_{l,k,s}(t) = b_{l,k}(t) \cdot P_k^{in}(t) \cdot \delta_{k,s}(t)$ , Eq. (13) can be equivalently transformed into the following constraints, according to Property 2 in Appendix.

$$P_{l,k,s}^{bo} = \sum_{t \in T} \omega_{l,k,s}(t), \quad \forall l \in \bar{N}, k < s \in \bar{S} \text{ or } \forall l \in \underline{N}, k < s \in \underline{S}. \quad (40)$$

$$\begin{cases} 0 \leq \omega_{l,k,s}(t) \leq M \cdot b_{l,k}(t), \\ \omega_{l,k,s}(t) \leq P_k^{in}(t) \cdot \delta_{k,s}(t), \\ \omega_{l,k,s}(t) \geq P_k^{in}(t) \cdot \delta_{k,s}(t) - M \cdot (1 - b_{l,k}(t)), \end{cases}$$

$$\forall l \in \bar{N}, k < s \in \bar{S}, t \in T \text{ or } \forall l \in \underline{N}, k < s \in \underline{S}, t \in T. \quad (41)$$

**Lemma 3.2.** In Eq. (34),  $(1 - \beta_l) \cdot y_{l,j}, z_{l,2|S|}(t - t_{l,2|S|}^w - t^{(re)}), (1 - \alpha_{l'})$  and  $z_{l',1}(t + t^{(le)})$  are all binary variables. According to Property 1 in Appendix, Eq. (34) can be linearized into the equivalent linear constraints by introducing the following four auxiliary variables  $\phi_{l,j} = (1 - \beta_l) \cdot y_{l,j}$ ,  $\psi_{l,j}(t) = \phi_{l,j} \cdot z_{l,2|S|}(t - t_{l,2|S|}^w - t^{(re)})$ ,  $\bar{\pi}_{l',j} = (1 - \alpha_{l'}) \cdot y_{l',j}$  and  $\bar{\zeta}_{l',j}(t) = \bar{\pi}_{l',j} \cdot z_{l',1}(t + t^{(le)})$ . That is,

$$\bar{C}N_j + \sum_{l \in \bar{N}} \psi_{l,j}(t) - \sum_{l' \in \bar{N}} \bar{\zeta}_{l',j}(t) \geq 0, \quad \forall j \in J, t, t - t_{l,2|S|}^w - t^{(re)}, t + t^{(le)} \in T. \quad (42)$$

$$\begin{cases} \phi_{l,j} \leq 1 - \beta_l, \\ \phi_{l,j} \leq y_{l,j}, \\ \phi_{l,j} \geq y_{l,j} - \beta_l, \\ \psi_{l,j}(t) \leq \phi_{l,j}, \\ \psi_{l,j}(t) \leq z_{l,2|S|}(t - t_{l,2|S|}^w - t^{(re)}), \\ \psi_{l,j}(t) \geq \phi_{l,j} + z_{l,2|S|}(t - t_{l,2|S|}^w - t^{(re)}) - 1, \\ \bar{\pi}_{l',j} \leq 1 - \alpha_{l'}, \\ \bar{\pi}_{l',j} \leq y_{l',j}, \\ \bar{\pi}_{l',j} \geq y_{l',j} - \alpha_{l'}, \\ \bar{\zeta}_{l',j}(t) \leq \bar{\pi}_{l',j}, \\ \bar{\zeta}_{l',j}(t) \leq z_{l',1}(t + t^{(le)}), \\ \bar{\zeta}_{l',j}(t) \geq \bar{\pi}_{l',j} + z_{l',1}(t + t^{(le)}) - 1, \end{cases}$$

$$\forall l \in \bar{N}, l' \in \bar{N}, j \in J, \quad t, t - t_{l,2|S|}^w - t^{(re)}, t + t^{(le)} \in T. \quad (43)$$

Similarly, Eq. (35) can be equivalently transformed into the following constraints by introducing four auxiliary variable  $\bar{\phi}_{l',j} = (1 - \beta_{l'}) \cdot y_{l',j}$ ,  $\bar{\psi}_{l',j}(t) = \bar{\phi}_{l',j} \cdot z_{l',1|S|}(t - t_{l',1|S|}^w - t^{(re)})$ ,  $\bar{\pi}_{l,j} = (1 - \alpha_l) \cdot y_{l,j}$  and  $\bar{\zeta}_{l,j}(t) = \bar{\pi}_{l,j} \cdot z_{l,|S|+1}(t + t^{(le)})$ . That is,

$$\bar{C}N_j + \sum_{l' \in \bar{N}} \bar{\psi}_{l',j}(t) - \sum_{l \in \bar{N}} \bar{\zeta}_{l,j}(t) \geq 0, \quad \forall j \in J, t, t - t_{l',1|S|}^w - t^{(re)}, t + t^{(le)} \in T. \quad (44)$$

$$\begin{cases}
\bar{\phi}_{l',j} \leq 1 - \beta_{l'}, \\
\bar{\phi}_{l',j} \leq y_{l',j}, \\
\bar{\phi}_{l',j} \geq y_{l',j} - \beta_{l'}, \\
\bar{\psi}_{l',j}(t) \leq \bar{\phi}_{l',j}, \\
\bar{\psi}_{l',j}(t) \leq z_{l',|S|}(t - t_{l',|S|}^w - \bar{t}^{(re)}), \\
\bar{\psi}_{l',j}(t) \geq \bar{\phi}_{l',j} + z_{l',|S|}(t - t_{l',|S|}^w - \bar{t}^{(re)}) - 1, \\
\pi_{l,j} \leq 1 - \alpha_l, \\
\pi_{l,j} \leq y_{l,j}, \\
\pi_{l,j} \geq y_{l,j} - \alpha_l, \\
\zeta_{l,j}(t) \leq \pi_{l,j}, \\
\zeta_{l,j}(t) \leq z_{l,|S|+1}(t + \underline{t}^{(le)}), \\
\zeta_{l,j}(t) \geq \pi_{l,j} + z_{l,|S|+1}(t + \underline{t}^{(le)}) - 1,
\end{cases}$$

$$\forall l \in \underline{N}, l' \in \overline{N}, j \in J, \quad t, t - t_{l',|S|}^w - \bar{t}^{(re)}, t + \underline{t}^{(le)} \in T. \quad (45)$$

### 3.4. Optimization model

According to Eq. (1)~Eq. (45), the joint optimization model for the train schedule, passenger flow control strategy and rolling stock circulation plan with multiple types of rolling stocks can be formulated as follows:

$$\begin{cases}
\min F = \lambda_1 \cdot \sum_{i \in T} \sum_{k \in \underline{S} \cup \underline{S}} P_k^{wa}(t) \cdot t_{unit} + \lambda_2 \cdot \left( \sum_{j \in J} \sum_{l \in \overline{N} \cup \underline{N}} y_{l,j} \cdot M_j^{cost} + \sum_{l \in \overline{N} \cup \underline{N}} [(1 - \alpha_l) \cdot M^{out}] \right) \\
s.t. \quad \text{constraints (1)–(12), (14)–(33), (40)–(45)}.
\end{cases} \quad (46)$$

**Remark 3.1.** It is worthy to mention that in the proposed method, we particularly introduce a binary decision variable  $y_{l,j}$  to indicate the type of rolling stocks for executing service  $l$  and establish some related constraints to make sure the feasibility of train operations. If we fix the values of decision variable  $y_{l,j}$  for each service  $l$ , the formulated model can be degenerated into the general model with only operating one type of rolling stocks. It means that our proposed model is a more general formulation and the traditional method with operating one type of rolling stocks is a special case of our proposed methods.

### 3.5. Complexity analyses

In the proposed model, the decision variables can be categorized into three classes. The first class is binary decision variables, including the indicators of a train entering the depot, exiting from the depot and successful connecting operations (i.e.,  $\alpha_l, \beta_l, \gamma_{l,l'}$ ), and the indicators of type of rolling stocks (i.e.,  $y_{l,j}$ ). The second class is integer decision variables (i.e.,  $t_{l,k}^a, t_{l,k}^d$ ) that represent the arrival and departure times of each service at each station. The last class is nonnegative decision variables (i.e.,  $P_{l,k}^{in}(t)$ ) related to the passenger flow control strategy. Since all the constraints are all formulated or reformulated as linear forms, this model is essentially a large-scale mixed-integer linear programming model with multiple big- $M$  constraints. The following discussion focuses on specifying the complexity of the proposed formulation in detail, such as the total numbers of variables and critical constraints involved in the problem, as shown in Table 3.

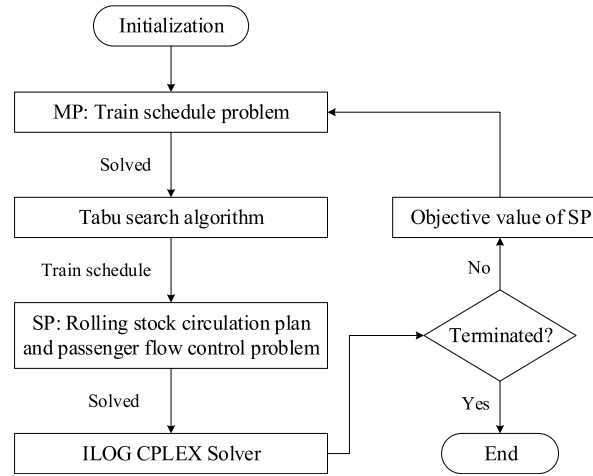
Clearly, the complexity of the proposed model is mainly determined by the numbers of services (i.e.,  $|\overline{N}|, |\underline{N}|$ ), stations (i.e.,  $|S|$ ), types of rolling stocks (i.e.,  $|J|$ ) and considered timestamps (i.e.,  $q$ ). To illustrate this problem clearly, an example with 20 services in each direction, 2 types of rolling stocks, 10 stations and 120 timestamps is given to identify the total number of decision variables. In this case, there are 960 binary variables with respect to  $\alpha_l, \beta_l, \gamma_{l,l'}$  and  $y_{l,j}$ , 800 integer variables with respect to  $t_{l,k}^a, t_{l,k}^d$  and 2400 continuous variables with respect to  $P_k^{in}(t)$ . While the number of auxiliary variables (i.e.,  $z_{l,k}(t), b_{l,k}(t)$ ) is 96 000. Clearly, the number of auxiliary variables substantially increases the scale of this example.

## 4. Solution approach

On the basis of the analyses above, the proposed model is reformulated as a mixed-integer linear programming model. In fact, if the problem is in a small scale, the proposed model can be efficiently solved by some optimization solvers (e.g., CPLEX, GORUBI). However, when the scale of the problem is large in the real-world situations, it is difficult to solve the proposed model since the model involves a large volume of integer, binary and continuous variables that make the model computationally intensive. To obtain a high-quality solution in a short computing time, we develop an effective heuristic algorithm in the following discussion. In detail, we first decompose the joint optimization model into two subproblems; then, we solve the subproblems with a heuristic algorithm. The details of the designed solution method will be elaborated below.

**Table 3**  
Numbers of variables and constraints in the formulation.

| Variable or constraints                        | Total number at most                                                     |
|------------------------------------------------|--------------------------------------------------------------------------|
| Binary variable $\alpha_i, \beta_i$            | $ \overline{N}  +  \underline{N} $                                       |
| Binary variable $\gamma_{i,i'}$                | $2 \cdot ( \overline{N}  \cdot  \underline{N} )$                         |
| Binary variable $y_{i,j}$                      | $ J  \cdot ( \overline{N}  +  \underline{N} )$                           |
| Integer variable $t_{i,k}^a, t_{i,k}^d$        | $ \overline{N}  \cdot  S  +  \underline{N}  \cdot  S $                   |
| Continuous variable $P_k^{in}(t)$              | $2 \cdot q \cdot  S $                                                    |
| Train schedule constraints(5)–(7)              | $3 \cdot q \cdot ( \overline{N}  \cdot  S  +  \underline{N}  \cdot  S )$ |
| Passenger flow control constraints(10)–(11)    | $4 \cdot q \cdot  S $                                                    |
| Passenger flow control constraints(14)–(17)    | $4 \cdot ( \overline{N}  \cdot  S  +  \underline{N}  \cdot  S )$         |
| Rolling stock circulation constraints(30)–(33) | $4 \cdot ( \overline{N}  \cdot  \underline{N} )$                         |



**Fig. 5.** The proposed algorithm.

#### 4.1. Decomposition of the proposed model

In model (46), the rolling stock circulation and passenger flow control are jointly optimized with the train schedule. In the solution process, the computational intensity comes from the coupling constraints among different components. To accelerate the search process, we decompose the joint optimization model into two parts, i.e., a main problem (MP) and a subproblem (SP), in which the main problem refers to the train scheduling process and the subproblem consists of rolling stock circulation planning and passenger flow control. Note that the decision variables in the train scheduling problem are related to the headway selection between different adjacent trains. We design a tabu search algorithm (TS) to search the feasible train schedules. Then, the feasible schedule is taken as the input for the subproblem to produce the corresponding optimal rolling stock circulation plan and passenger flow control strategy, which are also used to evaluate the generated schedule in the main problem. As the subproblem is linear, it can be efficiently solved by the CPLEX solver. For clarity, the structure of the proposed algorithm is presented in Fig. 5. This decomposition method has also been applied to the similar train schedule optimization problems, e.g., Shi et al. (2018), Qi et al. (2019), and Wu et al. (2021).

#### 4.2. Heuristic algorithm design

Next, we elaborate the designed tabu search algorithm for solving the main problem, including representation of the solution, initial solution generation, neighborhood structure, tabu list, tabu object and search process. The solution information is embedded into the subproblem to solve other components of the considered problem and to evaluate the generated train schedules.

**1. Representation of the solution.** In this paper, the running time of each service over each section and the dwell time at each station are assumed to be fixed. If the arrival time of a service at the origin station is given, we can deduce the arrival and departure times of each service at each station. Then, with the given arrival time of the first service at the origin station, the involved decision variable can be considered as the headway between two consecutive trains arriving at the origin station in the MP. Thus, this study takes a  $(|\overline{N}| + |\underline{N}| - 2)$ -dimensional headway array to represent a solution for the MP, and the corresponding train schedule can be generated by considering constraints (1)–(9).

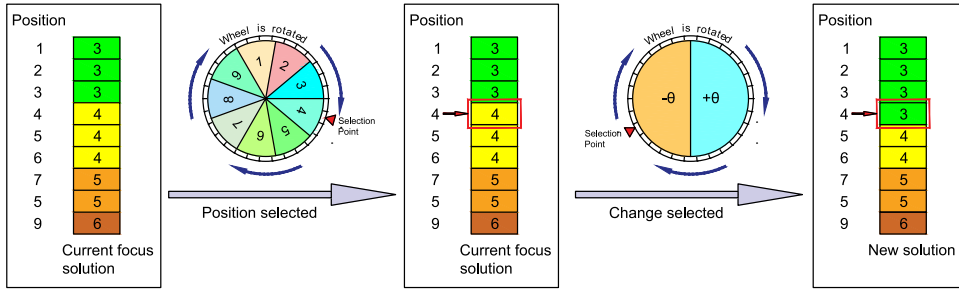


Fig. 6. The procedure of generating a new neighbor.

**2. Initial solution generation.** In this study, the initial solution is randomly generated in the first iteration of the tabu search process. That is, we first generate a total of  $(|\overline{N}| + |\underline{N}| - 2)$  headway variables randomly in the given range  $[h_{min}, h_{max}]$  for solution  $\hat{H}$ , where the feasibility of the generated solution is checked by Algorithm 1 given below. If this solution is feasible for all schedule-related constraints, we set the initial solution  $H^0 = \hat{H}$ ; otherwise, the solution must be regenerated until it is feasible. Then, the initial feasible solution is used as the input data for the SP, and the corresponding objective value solved by the CPLEX solver is regarded as an evaluation of this feasible train schedule.

---

**Algorithm 1** The procedure for checking the feasibility of a solution.

---

**Input:** A solution that needs to be checked

**Output:** The feasibility of the solution

**Step 1:** According to the headway constraints (8)–(9) and the input solution, calculate the arrival times of the last services at their origin stations for both directions. If  $t_{|\overline{N}|,1}^a$  and  $t_{|\underline{N}|,|S|+1}^a$  satisfy the constraints (3)–(4), go to Step 2; otherwise, the solution is infeasible, stop.

**Step 2:** According to the arrival and departure time constraints (1)–(2), calculate the departure times of the last services at each station, if  $t_{|\overline{N}|,k}^d, t_{|\underline{N}|,k}^d \in T$ , go to Step 3; otherwise, the solution is infeasible, stop.

**Step 3:** The solution is feasible, stop.

---

**3. Neighborhood structure.** The tabu search algorithm is an extension of the local search algorithm. In the searching process, we need to generate a series of solutions (i.e., neighbors) from the neighborhood of the current solution, and a new solution will be produced for the next iteration. To this end, we use a parameter  $\theta$  to represent the change of the current solution for the selected element to generate a new neighbor. For clarity, an example in Fig. 6 illustrates the generation of a new solution. Let  $\{3, 3, 3, 4, 4, 4, 5, 5, 6\}$  be a headway array for 10 services. Each element has the same probability of being selected and randomly minus or plus  $\theta$ . In this example, if the first element “4” is selected and  $-\theta$  ( $\theta = 1$ ) is selected as the change, a new solution  $\{3, 3, 3, 3, 4, 4, 5, 5, 6\}$  will be generated. If the newly generated neighbor is feasible, it can be inserted into the candidate neighbors; otherwise, a new neighbor should be regenerated until its feasibility is guaranteed. The detailed procedure of neighborhood generation is given in Algorithm 2.

---

**Algorithm 2** The procedure of neighborhood generation.

---

**Input:** Current focus solution:  $H^p$ , size of neighborhood:  $N^{nei}$

**Output:** Candidate neighbors of  $H^p$

**Step 1:** Generate a random integer  $r$  in the range of  $[1, solution.length]$  to represent the selected position of the current focus solution  $H^p$ , go to Step 2.

**Step 2:** Check the value  $\hat{v}$  of the selected position, if  $\hat{v} = h_{min}$ , let  $\hat{v} = h_{min} + \theta$ ; else if  $\hat{v} = h_{max}$ , let  $\hat{v} = h_{max} - \theta$ ; otherwise, let  $\hat{v} = \hat{v} + \theta$  or  $\hat{v} = \hat{v} - \theta$  with the same probability.

**Step 3:** Check the feasibility of the generated neighbor by Algorithm 1, if it is feasible, let index  $c = c + 1$  and go to step 4; otherwise, go to step 1.

**Step 4:** If  $c < N^{nei}$ , go to step 2; otherwise, output the candidate neighbors of the current focus solution, stop.

---

**4. Tabu list and tabu object.** To avoid trapping in local optimum solution in the search process, a tabu list is used to record the ever-focused solutions at a pre-specified number of nearest iterations. These solutions, which are also called tabu objects, can no longer be selected within a certain number of iterations (e.g., the tabu length). Three methods are usually used to determine which object will be put into the tabu list. (1) Tabu object is the value of the objective function: the corresponding solution with the tabu objective value cannot be taken as a search point before being released. For instance, if the tabued object is the



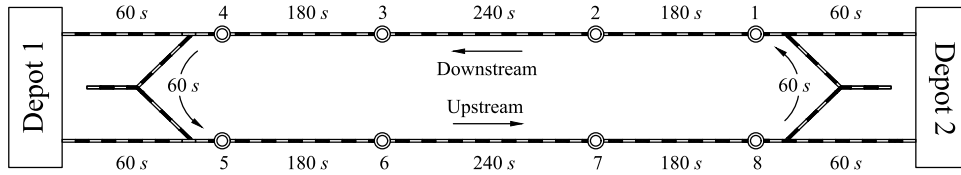


Fig. 7. The structure of the bidirectional metro corridor.

objective value with 100, then the solution with an objective of 100 cannot be selected as the current solution for the next iteration before being released. (2) Tabu object is the solution. When the tabued solution is the best local solution in the neighborhood, this solution cannot be selected, and a suboptimal solution will be selected. For example, if the neighborhood of the current solution is  $\{\{3, 3, 3, 4, 4, 4, 5, 5, 6\}, \{3, 3, 3, 4, 4, 4, 5, 5, 5\}, \{3, 3, 4, 4, 4, 4, 5, 5, 6\}, \{3, 4, 3, 4, 4, 4, 5, 5, 6\}\}$  with objective  $\{100, 105, 115, 120\}$ , while the best neighbor  $\{3, 3, 3, 4, 4, 4, 5, 5, 6\}$  is the tabu object, the suboptimal solution  $\{3, 3, 3, 4, 4, 4, 5, 5, 5\}$  will be selected for the next iteration. (3) Tabu object is the movement of the solution. The rationale of this method is to prohibit the solution by moving backwards, e.g., the process of moving element “4” to element “3” in Fig. 6. What is more, in order to increase the possibility of obtaining high-quality solution, if the objective value of the neighbor generated by the tabu object is better than the best objective value encountered up to now, the solution can be aspirated and used as the current solution in the next iteration.

**5. Algorithm procedure.** With the aforementioned specific techniques, the detailed procedure of the “TS + CPLEX” algorithm is summarized in Algorithm 3. The search process ends with two criteria. (1) If the present iteration is greater than a pre-specified maximum number of iterations  $M$ , the search process is terminated. We use  $S^p$  to represent the current iteration number. (2) If the objective value of the best solution does not update after a maximum number of iterations  $B_{max}$ , the algorithm can be terminated. We use  $S^b$  to denote the number of iterations for which the solution has not been updated.

---

**Algorithm 3** The procedure of the “TS + CPLEX” algorithm.

---

**Input:** Tabu length:  $L^{tabu}$ , size of neighborhood:  $N^{nei}$ , termination condition:  $M$  and  $B_{max}$

**Output:** Best solution  $H^*$ , best objective value  $V^*$

**Step 1:** Initialize tabu list and generate a feasible initial solution  $H^0$ . Calculate the objective value  $V^0$  of the initial solution, let current focus solution  $H^p = H^0$ , current objective value  $V^p = V^0$ ,  $H^* = H^p$ ,  $V^* = V^p$ , set  $S^p = 0$ ,  $S^b = 0$ , go to Step 2.

**Step 2:** Generate neighborhood  $N(H^p)$  of  $H^p$  by Algorithm 2, go to step 3.

**Step 3:** For each neighbor  $H(N^i)$ , calculate train timetable variables  $t_{l,k}^a, t_{l,k}^d$  in the MP; input the train timetable to the SP, solve the SP by CPLEX solver, and obtain the corresponding objective value  $V(N^i)$ ; go to step 4.

**Step 4:** Rank the neighbors in  $N(H^p)$  and find the non-tabu best or aspired solution  $H(N^*)$  in  $N(H^p)$ . Let  $H^p = H(N^*)$ ,  $V^p = V(N^*)$ , set  $S^p = S^p + 1$ , go to step 5.

**Step 5:** Update the tabu list and the best solution. If  $V^* > V^p$ , let  $H^* = H^p$ ,  $V^* = V^p$ , set  $S^b = 0$ ; otherwise, set  $S^b = S^b + 1$ , go to step 6.

**Step 6:** Check the termination condition. If  $S^p < M$  and  $S^b < B_{max}$ , go to step 2; otherwise, output the best solution  $H^*$  and the best objective value  $V^*$ , stop.

---

## 5. Numerical experiments

In this section, a series of numerical experiments based on a small-scale example and the real-world Beijing metro Batong line are conducted to demonstrate the performance and efficiency of our proposed methods, in which Visual Studio C++ 2019 with IBM ILOG CPLEX 12.9 solver is employed to code the searching process for producing an optimal solution. All the experiments are implemented on a Windows 10 personal computer with Intel Core i5-10500 CPU and 16G RAM.

### 5.1. A small-scale example

In this example, we consider a simple bidirectional metro corridor with 4 stations in each direction, i.e.,  $\underline{S} = \{1, 2, 3, 4\}$ ,  $\bar{S} = \{5, 6, 7, 8\}$ , and two depots respectively at the end of the corridor, as shown in Fig. 7. There are three sections in each direction, and the travel times of each train over each section are assumed to be 180 s, 240 s, 180 s, respectively. The dwell time of each service at each station is set as 60 s. The considered planning time horizon is 60 min, with  $t_{unit} = 1$  min. A total of 20 services (i.e., 10 services in each direction) are considered in the experiments. In addition, two types of rolling stocks are available for different services, and the capacities corresponding to these two types of rolling stocks are respectively set as 650 (persons) and 400 (persons). Moreover, the fixed costs of executing the service by different types of rolling stocks are respectively 3200 and 1600, and the movement costs of a rolling stock exiting from the depots is set as 1000. The values of other parameters are shown in Table 4.

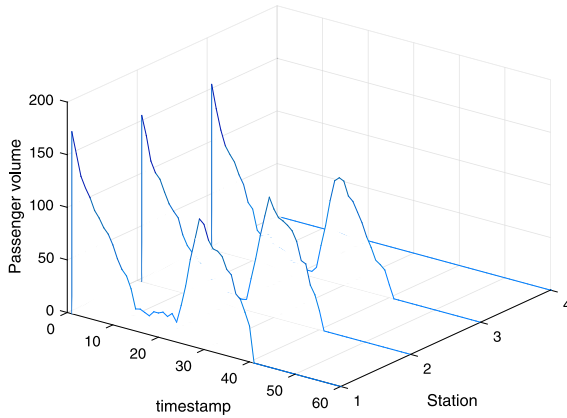
The associated passenger demands for each direction are shown in Fig. 8 and Table 5, which demonstrate the nonequilibrium distribution of passengers in the two directions. The passenger flow in the upstream direction obeys a normal distribution, where

**Table 4**  
Relevant parameters in the small-scale example.

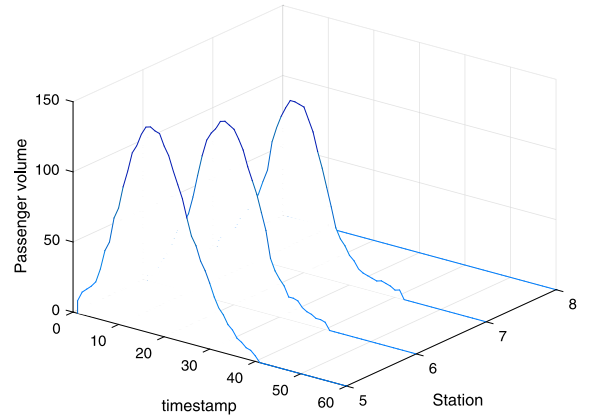
| Parameters                                                           | Notations                          | Values |
|----------------------------------------------------------------------|------------------------------------|--------|
| Minimum headway                                                      | $h_{min}$                          | 180 s  |
| Maximum headway                                                      | $h_{max}$                          | 360 s  |
| Turn-around time                                                     | $T^{roll}$                         | 60 s   |
| Maximum turn-around time                                             | $t_{max}^{(roll)}$                 | 240 s  |
| The time of the rolling stock returning to the depot                 | $t_{(re)}^{(re)}, t_{(le)}^{(le)}$ | 60 s   |
| The time of the rolling stock exiting from the depot                 | $t_{(le)}^{(le)}, t_{(re)}^{(re)}$ | 60 s   |
| Inventory trains in depot 1                                          | $CN_1, CN_2$                       | 4, 5   |
| Inventory trains in depot 2                                          | $CN_1, CN_2$                       | 5, 4   |
| The arrival timestamp of the first service at origin station         | $t_{(le)}^{(fi)}, t_{(re)}^{(fi)}$ | 1      |
| The earliest arrival timestamp of the last service at origin station | $t_{(le)}^{(la)}, t_{(re)}^{(la)}$ | 39     |

**Table 5**  
Passenger demands in the small-scale example.

| Station index | Total passengers | Station index | Total passengers |
|---------------|------------------|---------------|------------------|
| 1             | 2846             | 5             | 2758             |
| 2             | 2418             | 6             | 2243             |
| 3             | 2121             | 7             | 1962             |
| 4             | 0                | 8             | 0                |



(a) Passenger demand in downstream direction.



(b) Passenger demand in upstream direction.

**Fig. 8.** Variation of passenger demand in the small-scale example.

**Table 6**  
The values of  $\delta_{k,s}(t)$  in the downstream direction.

| (a) Timestamps 1-15 |     |     |     |     | (b) Timestamps 16-30 |     |     |     |     | (c) Timestamps 31-60 |     |     |     |     |
|---------------------|-----|-----|-----|-----|----------------------|-----|-----|-----|-----|----------------------|-----|-----|-----|-----|
| $k$                 | $s$ |     |     |     | $k$                  | $s$ |     |     |     | $k$                  | $s$ |     |     |     |
|                     | 1   | 2   | 3   | 4   |                      | 1   | 2   | 3   | 4   |                      | 1   | 2   | 3   | 4   |
| 1                   | 0   | 0.1 | 0.4 | 0.5 | 1                    | 0   | 0.2 | 0.3 | 0.5 | 1                    | 0   | 0.1 | 0.3 | 0.6 |
| 2                   | 0   | 0   | 0.5 | 0.5 | 2                    | 0   | 0   | 0.4 | 0.6 | 2                    | 0   | 0   | 0.5 | 0.5 |
| 3                   | 0   | 0   | 0   | 1   | 3                    | 0   | 0   | 0   | 1   | 3                    | 0   | 0   | 0   | 1   |
| 4                   | 0   | 0   | 0   | 0   | 4                    | 0   | 0   | 0   | 0   | 4                    | 0   | 0   | 0   | 0   |

the peak period is in time interval [11, 25]. The peak period in the downstream direction is in time intervals [1, 8] and [28, 34]. In addition, the entire time horizon is discretized into three time intervals, i.e., [1, 15], [16, 30] and [31, 60]. In each time interval, the ratio of passengers with destination station  $k$  in the boarding passengers at station  $s$ , i.e.,  $\delta_{k,s}(t)$ , is assumed to be equal, as shown in Tables 6 and 7.

According to the above experimental data, we design different instances to assess the performance of the proposed approaches, including the weight coefficient analysis of the objective function, solution analysis of the proposed model, and performance comparison of different approaches.

**Table 7**The values of  $\delta_{k,s}(t)$  in the upstream direction.

| (a) Timestamps 1-15 |     |     |     |     | (b) Timestamps 16-30 |     |     |     |     | (c) Timestamps 31-60 |     |     |     |     |
|---------------------|-----|-----|-----|-----|----------------------|-----|-----|-----|-----|----------------------|-----|-----|-----|-----|
| $k$                 | $s$ |     |     |     | $k$                  | $s$ |     |     |     | $k$                  | $s$ |     |     |     |
|                     | 5   | 6   | 7   | 8   |                      | 5   | 6   | 7   | 8   |                      | 5   | 6   | 7   | 8   |
| 5                   | 0   | 0.1 | 0.4 | 0.5 | 5                    | 0   | 0.2 | 0.4 | 0.4 | 5                    | 0   | 0.1 | 0.3 | 0.6 |
| 6                   | 0   | 0   | 0.1 | 0.9 | 6                    | 0   | 0   | 0.3 | 0.7 | 6                    | 0   | 0   | 0.5 | 0.5 |
| 7                   | 0   | 0   | 0   | 1   | 7                    | 0   | 0   | 0   | 1   | 7                    | 0   | 0   | 0   | 1   |
| 8                   | 0   | 0   | 0   | 0   | 8                    | 0   | 0   | 0   | 0   | 8                    | 0   | 0   | 0   | 0   |

**Table 8**

Computation results for different weights in the objective function.

| Instance index | Objective value | $WT/min$ | $C$   | $HT$ | $LT$ | $CT/s$ | $GAP$ |
|----------------|-----------------|----------|-------|------|------|--------|-------|
| K-0.1-0.9      | 52837           | 53174    | 52800 | 8    | 12   | 3011.2 | 2%    |
| K-0.3-0.7      | 50480           | 33866    | 57600 | 11   | 9    | 2243.5 | 2%    |
| K-0.5-0.5      | 44633           | 28065    | 61200 | 12   | 8    | 1040.2 | 2%    |
| K-0.7-0.3      | 37929           | 27699    | 61800 | 13   | 7    | 886.4  | 2%    |
| K-0.9-0.1      | 31106           | 27584    | 62800 | 13   | 7    | 297.7  | 2%    |

### (i) Weight coefficient analysis of the objective function

First, we explore the performance of the proposed model with different weight coefficients in the objective function (39). For clarity, an index (i.e.,  $K-\lambda_1-\lambda_2$ ) is introduced to represent the integrated weight coefficients of two components in the objective function. For example, K-0.1-0.9 indicates  $\lambda_1 = 0.1$  and  $\lambda_2 = 0.9$ . The CPLEX solver is adopted to solve the approximate optimal solutions of the designed 5 instances, in which the termination condition is set as the relative duality gap less than 2%. The detailed computational results are displayed in Table 8, where some critical indexes are listed, including the passenger waiting time ( $WT$ ), the operating costs ( $C$ , i.e., the fixed costs of services  $CR$  plus the movement costs from the depot  $CS$ ), the total number of high-capacity train services ( $HT$ ), the total number of low-capacity train services ( $LT$ ), the computational time ( $CT$ ), and the relative gap returned by CPLEX ( $GAP$ ).

It follows from Table 8 that although the CPLEX solver can return an approximate optimal solution with relative duality gap of 2% for each instance, the setting of coefficient weights in the objective function substantially impacts the final solution of the proposed model. Clearly, among these instances, the minimum passenger waiting time and minimum operating costs are respectively 27,584 min and 52,800. The passenger waiting time shows an increase tendency with the increase of  $\lambda_2$  (or decrease of  $\lambda_1$ ), while the operating costs have an opposite variation tendency. These results imply that if we want to reduce the passenger waiting time, we can set  $\lambda_1$  as a much larger value, which needs to put more high-capacity rolling stocks into operations and unavoidably leads to higher operating costs for the companies. On the other hand, if we want to reduce the operating costs,  $\lambda_2$  should be set as a much larger value, which might result in an increase in the passenger waiting time. In practice, according to this tendency, the managers can make a trade-off between these two objectives by setting different values of  $\lambda_1, \lambda_2$  and generate a more appropriate train schedule and rolling stock circulation plan with the consideration of the passenger flow control strategy. In addition, as instance K-0.7-0.3 provides a good compromise between these two components, the following experiments will take the same weight coefficients if there is no further statement.

### (ii) Solution analysis for the proposed model

Next, we analyze the solution obtained with weight coefficients  $\lambda_1 = 0.7$  and  $\lambda_2 = 0.3$  in detail so as to show the effectiveness of the proposed methods. By the CPLEX solver, we obtain the near-optimal solution for this case in 886.4 s with 2% relative duality gap. The approximate optimal objective value is 37 929, corresponding to 27 699 min passenger waiting time, 52 800 fixed costs and 9000 movement costs. The optimal train schedule and rolling stock circulation plan are displayed in Fig. 9, in which the blue and red lines denote the detailed services in the downstream and upstream direction, respectively. In addition, the bars filled with orange and blue represent the accumulated passengers (i.e.,  $P_k^{wd}(t)$ ) in station halls in the downstream and upstream directions, respectively. To present the dynamic evolution of passengers, we use numbers in the rectangles to represent the volume of passengers in each service over different sections. Moreover, the solid and dashed lines, respectively, represent that the associated services are executed by the high-capacity and low-capacity rolling stocks. To display the rolling stock circulation plan in the generated train schedule clearly, we use red lines to denote the turn-around operations, and the red and blue rectangles represent the trains exiting from the depots and returning to the depots, respectively.

As shown in Fig. 9, to better match the provided services and passenger demand, the departure headway is shorter during peak periods since more passengers are involved, while it is relatively large during other time periods. Accordingly, more high-capacity rolling stocks will be put into operation during peak hours, while more low-capacity rolling stocks are used during off-peak hours. In terms of the rolling stock circulation plan, most of services can be connected via turn-around operations. For this example with 20 services and a 60 min time horizon, at least 8 rolling stocks are required to execute the planned train schedule. In addition, the number of accumulated passengers at each station shows various tendencies, i.e., different numbers of passengers can enter the platform. When the number of arrival passengers is large, more passengers are allowed to enter (e.g., station 5). However, the ratio of passengers entering the platform becomes small due to the passenger control strategy. Conversely, if there are fewer passengers

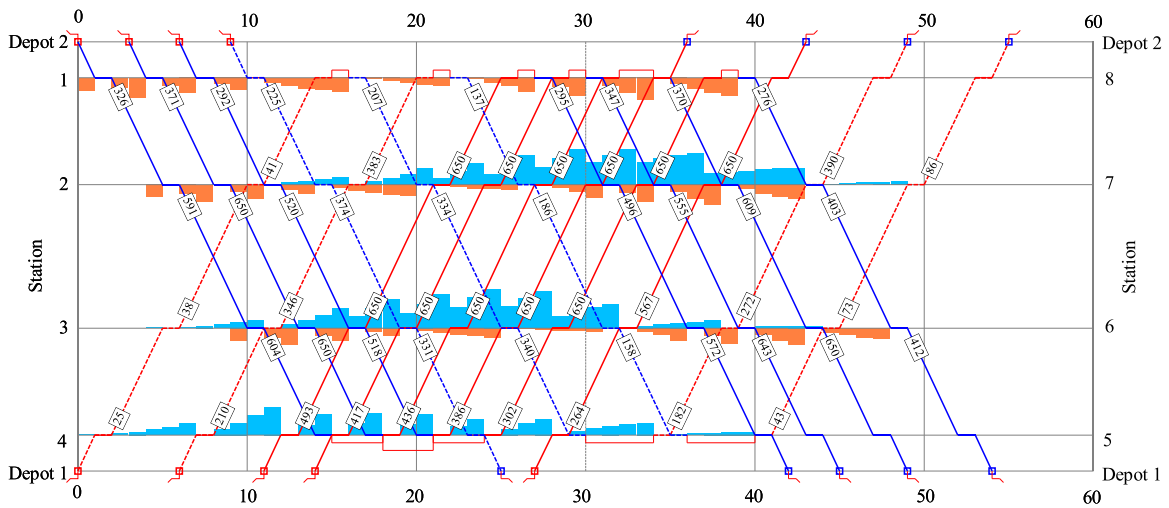


Fig. 9. The obtained train schedule and rolling stock circulation plan for the small-scale example.

**Table 9**  
Performance comparison for different instances with CPLEX and TS+CPLEX.

| Instance index | Algorithm | SN | Objective value | $WT/(min)$ | $C$   | CT/(s) | GAP   |
|----------------|-----------|----|-----------------|------------|-------|--------|-------|
| E-8-60         | CPLEX     | –  | 28708           | 20012      | 49000 | 9.4    | 2%    |
|                | TS+CPLEX  | 14 | 28708           | 20012      | 49000 | 43.5   | –     |
| E-9-60         | CPLEX     | –  | 33024           | 24120      | 53800 | 137.9  | 2%    |
|                | TS+CPLEX  | 16 | 33875           | 25079      | 54400 | 37.8   | –     |
| E-10-60        | CPLEX     | –  | 37929           | 27699      | 61800 | 886.4  | 2%    |
|                | TS+CPLEX  | 18 | 38245           | 28150      | 61800 | 54.3   | –     |
| E-11-60        | CPLEX     | –  | 39930           | 29015      | 65400 | 3899.1 | 2%    |
|                | TS+CPLEX  | 20 | 38724           | 25749      | 69000 | 73.0   | –     |
| E-12-80        | CPLEX     | –  | 47584           | 35234      | 76400 | 7200.0 | 36.2% |
|                | TS+CPLEX  | 22 | 45856           | 32765      | 76400 | 249.2  | –     |
| E-13-80        | CPLEX     | –  | 59387           | 46953      | 88400 | 7200.0 | 38.2% |
|                | TS+CPLEX  | 24 | 55983           | 44061      | 83800 | 220.1  | –     |

arriving, fewer passengers enter the station (e.g., stations 6 and 7). But, the ratio of passengers entering the platform is much larger since there are more space remaining for the passengers to enter, and almost all the arrival passengers can enter the platform. In general, there is no station where a large number of passengers are accumulated, which indicates that passengers are rigorously controlled at each station during peak hours.

### (iii) Performance comparison of two approaches for the proposed model

Note that although the problem with small-scales can be well solved by the CPLEX solver, the computational time is greatly influenced by the scale of the problem. To assess the performance of our designed algorithm, we use the “TS+CPLEX” to resolve this example. In this heuristic, the parameters are set as follows: the maximum number of iterations is  $M = 100$ ; the size of the neighborhood is  $N^{nei} = 18$ ; the maximum number of iterations without improving the best encountered solution is  $S^b = 10$ . With 54.3 s computation, the model is solved by the proposed algorithm. In comparison to the CPLEX, the computing time can be reduced up to 93.9% by our designed algorithm. Moreover, the objective value of the obtained near-optimal solution is 38,245, which is very close to that obtained by the CPLEX solver. In this sense, the “TS+CPLEX” algorithm can efficiently provide a high-quality solution within a very short time.

To further illustrate the performance of the developed “TS+CPLEX”, we design a set of experimental instances with different numbers of services and time horizons. For convenience, an index, i.e., E-A-T, is introduced to represent the instance information, in which  $A$  denotes the number of services in each direction and  $T$  denotes the length of the considered time horizon. For example, the instance E-10-60 indicates that 10 services in each direction need to be scheduled in time horizon  $[1, 60]$ . For consistency, the passenger demands and number of available rolling stocks are also adjusted according to the number of considered services and length of the time horizon. The termination condition of CPLEX solver is set as that the relative gap is less than 2% or the CPU time reaches 7200 s.

Table 9 displays the computation results for 6 instances with different numbers of services and time horizons. The corresponding results, including the size of the neighborhood (SN), passenger waiting time ( $WT$ ), operating costs ( $C$ ), computational time (CT), and relative gap obtained by CPLEX (GAP), are displayed in the table. Clearly, the “TS+CPLEX” algorithm needs much shorter computing time to obtain solutions that are very close to or better than those produced by CPLEX solver in these instances. In particular, for the small-scale instances, the solution quality is similar for these two approaches, while the computational time of CPLEX solver is less

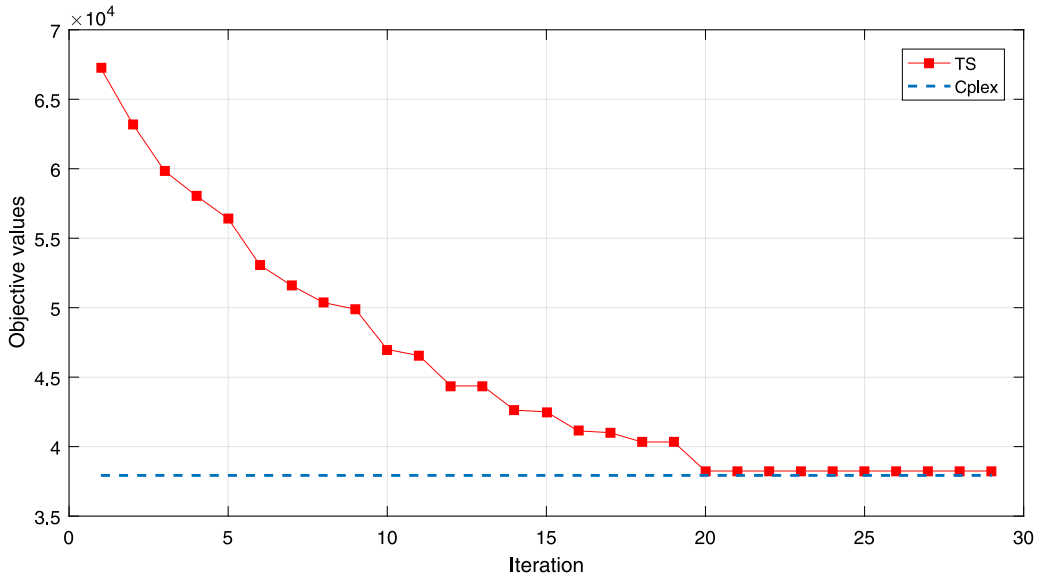


Fig. 10. The variation in the encountered best objective value at each iteration.

than that of the “TS+CPLEX” algorithm (e.g., instance E-8-60). With the number of services increasing, the solution outputted by these two approaches are fairly close, but the “TS+CPLEX” algorithm has a much faster convergence speed (e.g., instances E-9-60 and E-10-60). When we consider much larger instances with more services and longer time horizons, the computational efficiency and performance of “TS+CPLEX” are greatly better than those of the CPLEX solver (e.g., instances E-11-60, E-12-80 and E-13-80).

Next, we intend to analyze the variation of the best encountered objective value in each iteration in the searching process of the proposed algorithm. For illustrative convenience, Fig. 10 plots the variation tendency of the best objective values for instance E-10-60. Clearly, the developed “TS+CPLEX” algorithm iteratively improves the quality of the solution, and the near-optimal objective value is achieved after 29 iterations. In Fig. 10, the best encountered objective drastically decreases in the first several iterations since the initial solution is usually much far away from the best solution. In detail, from iterations 10 to 20, the best objective varies with a relative flat tendency, and in iteration 20, the searched best objective is very close to that obtained by CPLEX. Moreover, in the last 10 iterations, the objective almost cannot be further improved since it is already much close to the best solution. These computational results indicate that our proposed algorithm has a good performance for the proposed joint optimization model.

#### (iv) Performance comparison with different operation modes

To further validate the performance of our proposed methods, we further compute the experimental results without considering multiple types of rolling stocks. Specifically, we calculate the number of waiting passengers and operating costs considering only one type of rolling stocks for the train schedule obtained by the CPLEX in instance E-10-60. The following three instances will be compared: *case 1*: all services are undertaken by high-capacity rolling stocks; *case 2*: all services are undertaken by two types of rolling stocks (i.e., E-10-60); *case 3*: all services are implemented by low-capacity rolling stocks. After optimization, the passenger waiting time, fixed costs of services, movement costs in *case 1* are 27 699, 64 000 and 8000, respectively. Nevertheless, since all services are executed by low-capacity rolling stocks in *case 3*, a portion of the passengers cannot be transported due to the insufficient transportation capacity, which violates the passenger loading constraints. We thus record the passenger waiting time as a sufficiently large value. In addition, the fixed costs of services and movement costs in this case are 32 000 and 8000, respectively. Then, we compare these indexes with those of *case 2* obtained by the proposed method. Table 10 gives a detailed comparison of these three instances, in which  $dev^t$  and  $dev^c$  are respectively defined as follows

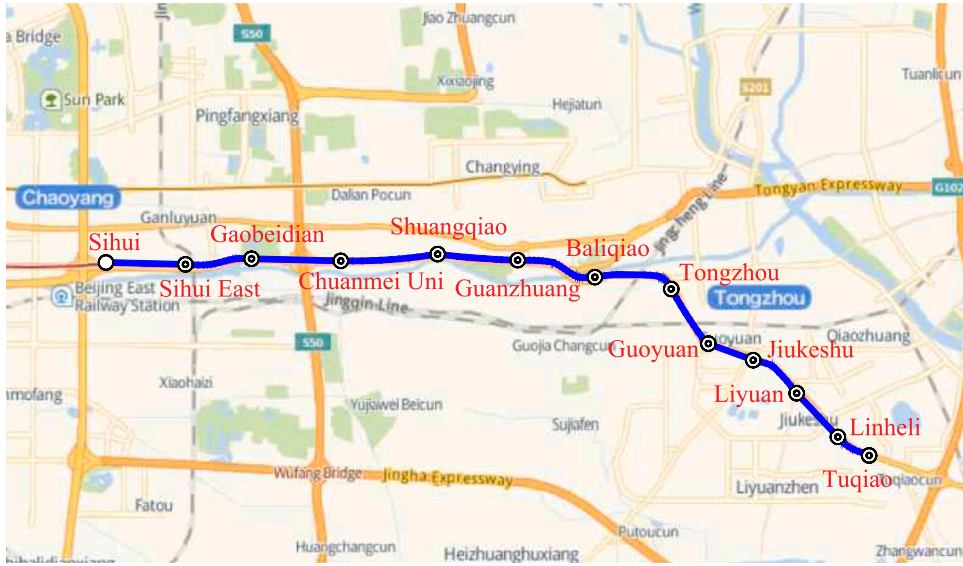
$$dev^t = \frac{WT \text{ in case 1} - WT \text{ in case 2}}{WT \text{ in case 1}},$$

$$dev^c = \frac{C \text{ in case 1} - C \text{ in case 2}}{C \text{ in case 1}}.$$

As we can see, in Table 10, since *case 1* uses large-capacity rolling stocks, this case corresponds to the largest operating costs. Additionally, *case 1* and *case 2* have the same value for passenger waiting time since the train schedule and passenger dynamic evolution are identical in these two instances, even with different operation modes. Although high-capacity rolling stocks can satisfy passenger demands, this mode unavoidably brings transportation capacity waste in off-peak hours. By properly allocating low-capacity rolling stocks in *case 2*, the service quality can be guaranteed with lower operating costs (decrease by 14.2%). Practically, it is not desirable to only use the low-capacity rolling stocks for reducing operating cost since the passenger demands cannot be satisfied due to the insufficient capacity. Thus, different types of rolling stocks can be used to match the provided services and passenger

**Table 10**  
Comparison of train operation indexes with different operation modes.

| Instance | WT / (min) | dev <sup>f</sup> | CR    | CS   | C     | dev <sup>c</sup> |
|----------|------------|------------------|-------|------|-------|------------------|
| Case 1   | 27699      | –                | 64000 | 8000 | 72000 | –                |
| Case 2   | 27699      | 0%               | 52800 | 9000 | 61800 | 14.2%            |
| Case 3   | ∞          | –                | 32000 | 8000 | 40000 | –                |



**Fig. 11.** Illustration of Beijing metro Batong line.  
Source: Gaode map.

demands in a much better manner, which can obtain a system optimized scheme for both metro companies and passengers. The relevant results also verify the effectiveness of the proposed methods.

## 5.2. Large-scale experiments on Beijing metro Batong line

To further test the performance of the proposed approaches for large-scale problems, some new numerical experiments are performed on a realistic metro line, i.e., Beijing metro Batong line, to collaboratively generate the optimized train schedule, rolling stock circulation plan and passenger flow control strategy. The Batong line has a total length of 18.94 km and connects a suburb area (i.e., Tongzhou district) and downtown of Beijing city. In the morning peak hours of a workday, a considerable number of commuters travel to their offices via this metro line and then return to home in the evening. Thus, the tidal characteristics of passenger demand is notable during the peak periods of daily operations, which is suitable for applying the proposed methods in this study. The metro line consists of 13 stations and 12 running sections, as shown in Fig. 11, where the upstream direction is from “Sihui” to “Tuqiao” and the downstream direction is from “Tuqiao” to “Sihui”. Two depots are respectively located at the two ends of the corridor.

### 5.2.1. Data preparation

Before conducting the numerical experiments, we first pre-generate the input data for the formulated model in these large-scale experiments, including station dwell time, section running time, and dynamic passenger demands. Most of the experimental data are extracted from historical recorded operation data on this metro line. The dwell time at each station and running time over each section are given in Tables 11 and 12, respectively.

In this set of experiments, the considered time horizon is 6:30–11:30 with  $t_{unit} = 1$  min, which consists of morning peak hours and partial of off-peak hours, as shown in Fig. 12. The dynamic passenger flow at each station is extracted from practical OD trips data of passengers collected on a weekday in June, 2016. As displayed in Fig. 12, the passenger demand in the downstream direction is much higher than that in the upstream direction, showing a typically tidal characteristic.

In the experiments, two types of rolling stocks with maximum capacities of 2000 (persons) and 1100 (persons) are respectively taken into considerations. The fixed costs of the service executed by two types of rolling stocks are 4000 and 1800, respectively; the movement cost of a rolling stock exiting from the depots is 2000. The total number of involved services in both directions is set as 55. The other relevant parameters are listed in Table 13.

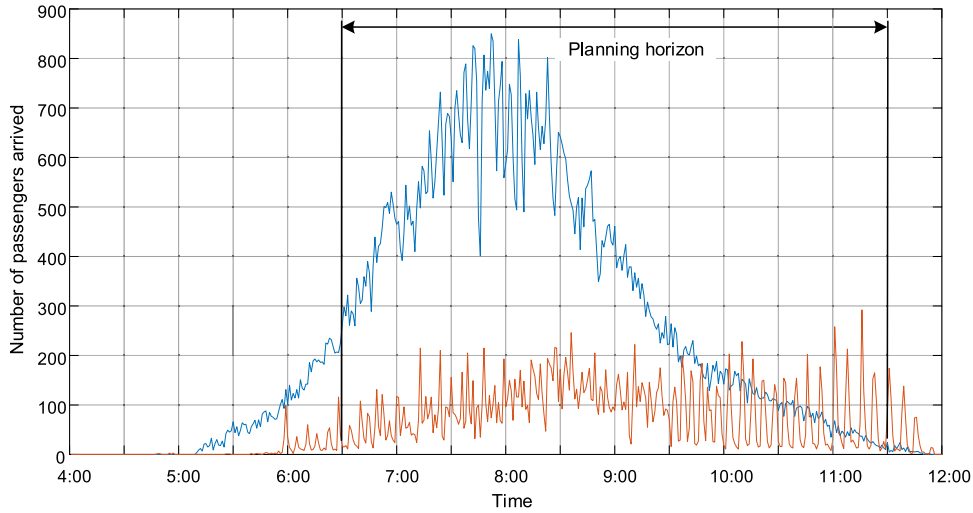


**Table 11**  
Station dwell time in the large-scale experiments.

| Station       | Dwell time/s | Station  | Dwell time/s |
|---------------|--------------|----------|--------------|
| Sihui         | 60           | Tongzhou | 60           |
| Sihui East    | 60           | Guoyuan  | 60           |
| Gaobeidian    | 60           | Jiukeshu | 60           |
| Chuangmei Uni | 60           | Liyuan   | 60           |
| Shuangqiao    | 60           | Linheli  | 60           |
| Guanzhuang    | 60           | Tuqiao   | 60           |
| Baliqiao      | 60           |          |              |

**Table 12**  
Section running time in the large-scale experiments.

| Section                    | Running time/s | Section                    | Running time/s |
|----------------------------|----------------|----------------------------|----------------|
| Sihui → Sihui East         | 180            | Sihui East → Sihui         | 180            |
| Sihui East → Gaobeidian    | 120            | Gaobeidian → Sihui East    | 120            |
| Gaobeidian → Chuangmei Uni | 180            | Chuangmei Uni → Gaobeidian | 180            |
| Chuangmei Uni → Shuangqiao | 180            | Shuangqiao → Chuangmei Uni | 180            |
| Shuangqiao → Guanzhuang    | 180            | Guanzhuang → Shuangqiao    | 180            |
| Guanzhuang → Baliqiao      | 180            | Baliqiao → Guanzhuang      | 180            |
| Baliqiao → Tongzhou        | 180            | Tongzhou → Baliqiao        | 180            |
| Tongzhou → Guoyuan         | 120            | Guoyuan → Tongzhou         | 120            |
| Guoyuan → Jiukeshu         | 120            | Jiukeshu → Guoyuan         | 120            |
| Jiukeshu → Liyuan          | 120            | Liyuan → Jiukeshu          | 120            |
| Liyuan → Linheli           | 120            | Linheli → Liyuan           | 120            |
| Linheli → Tuqiao           | 120            | Tuqiao → Linheli           | 120            |



**Fig. 12.** Real-world passenger flow data of Beijing metro Batong line.

### 5.2.2. Computational results

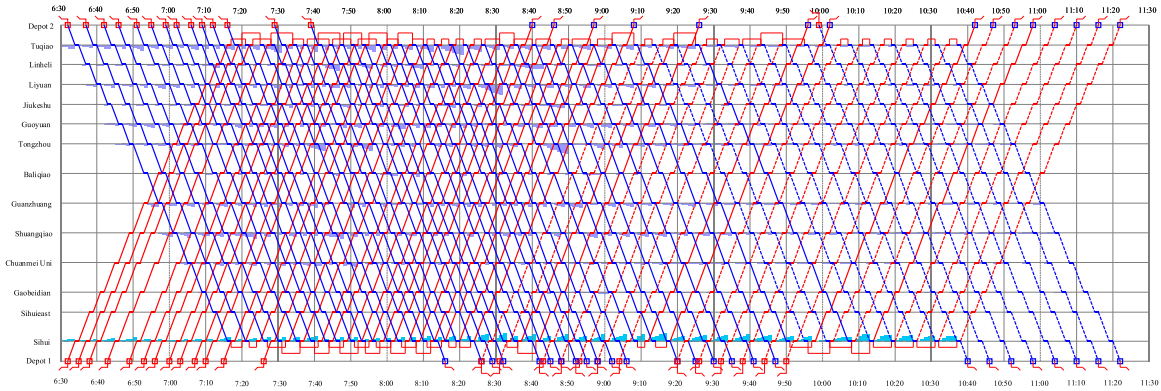
With the experimental data prepared above, we first use CPLEX solver to solve the developed joint optimization model with the maximum computing time of 2 h. With this time threshold, the CPLEX solver cannot even find a feasible solution due to its sensitivity to the scale of the problem. Alternatively, we next adopt the developed “TS+CPLEX” to solve the proposed model. The involved parameters for “TS+CPLEX” are given below. The tabu length is 10; the maximum number of iterations is  $M = 100$ ; the size of the neighborhood is  $N^{nei} = 80$ , and the maximum number of iterations without improving the best solution is  $S^b = 10$ . After a total of 82 iterations, the search algorithm is terminated with a CPU time of 18 305.1 s = 5.1 h. The objective value is 296 646; the passenger waiting time is 231,180 min; the corresponding fixed costs are 367 400; and the movement costs are 82 000.

Next, we analyze the results obtained by the “TS+CPLEX” algorithm explicitly. The obtained train schedule and passengers waiting in station halls at each timestamp with the passenger flow control strategy (denoted by PFCs) are shown in Fig. 13. The symbols in Fig. 13 are similar to those in the small-scale example, where the bars denote the accumulated passengers in the corresponding station halls (i.e.,  $P_k^{wd}(t)$ ). In the solution, two types of rolling stocks are assigned to execute the services in the optimized train schedule, where the full lines represent trajectories of high-capacity rolling stocks and the dashed lines represent trajectories of low-capacity rolling stocks.



**Table 13**  
Relevant parameters in the large-scale example.

| Parameters                                                      | Notations                            | Values  |
|-----------------------------------------------------------------|--------------------------------------|---------|
| Minimum headway                                                 | $h_{min}$                            | 3 min.  |
| Maximum headway                                                 | $h_{max}$                            | 6 min.  |
| Turn-around time                                                | $T^{roll}$                           | 2 min.  |
| Maximum turn-around time                                        | $t_{max}^{(roll)}$                   | 6 min   |
| The time of the rolling stock returning to the depot            | $t_{(re)}^{(re)}, t_{(re)}^{(le)}$   | 120 s.  |
| The time of the rolling stock exiting from the depot            | $t_{(le)}^{(le)}, t_{(le)}^{(re)}$   | 120 s.  |
| Inventory trains in depot 1                                     | $\overline{CN}_1, \overline{CN}_2$   | 13, 13. |
| Inventory trains in depot 2                                     | $\underline{CN}_1, \underline{CN}_2$ | 14, 2.  |
| The arrival time of the first service at origin station         | $t_{(fi)}^{(fi)}, t_{(fi)}^{(li)}$   | 4.      |
| The earliest arrival time of the last service at origin station | $t_{(la)}^{(la)}, t_{(la)}^{(li)}$   | 247.    |



**Fig. 13.** The obtained train schedule and rolling stock circulation plan with PFCS.

As shown in Fig. 13, to reduce the operating costs, different types of rolling stocks are arranged to execute the services at different periods. Specifically, high-capacity rolling stocks are scheduled to implement the services over time interval [6:30, 9:10] in the downstream direction due to the large number of passenger demands during this period. Then, most of the high-capacity rolling stocks return to the depots, and low-capacity rolling stocks are put into operations for the next off-peak periods (e.g., [9:10, 11:30]). Meanwhile, during the peak hours, the services in the downstream direction are scheduled with a relatively small headway. In comparison, the headway is more flexible in the upstream direction since the passenger demand is much smaller than that in the downstream direction, and the transportation capacity is sufficient to transport all the associated passengers. Thus, in terms of meeting passenger demand, the optimal headway in the upstream direction has more possibilities to improve the number of successful connected services. These results demonstrate the necessity of joint optimization of the train schedule and rolling stock circulation plan with consideration of the dynamic passenger demand and the operating costs.

As for the performance of passenger flow control strategy, it follows from Fig. 13 that many passengers are stranded in the station halls in the downstream direction due to the large number of passenger demand. Since the waiting passengers are distributed among different stations, the maximum number of waiting passengers at each station is no more than 800 persons. For comparison, on the basis of the same train schedule, we also obtain the results without passenger flow control strategy (PFCS). In this case, the passenger waiting time increases to 267,751 min, which indicates that the passenger flow control strategy can bring 13.6% reduction of the total passenger waiting time. For clarity, the accumulation situation of passengers without PFCS is shown in Fig. 14. It is easy to see that, the distribution of accumulated passengers becomes extremely unbalanced at several downstream stations, such as “Shuangqiao” and “Chuanmei Uni”. This is mainly due to that most of services are fully loaded when they arrive at these stations, and no sufficient capacity is left. For instance, there are 11 services that are full of passengers when they arrive at “Chuanmei Uni” station. Accordingly, the maximum number of waiting passengers at “Chuanmei Uni” is 821 persons, which is much higher than that with PFCS. In contrast, under the passenger flow control strategy, some of passengers are restricted from boarding the services at some stations (e.g., “Tuqiao”), and almost no services are fully loaded when they arrive at “Chuanmei Uni” station. Thus, a proper passenger flow control strategy can effectively reduce the potential risks caused by the passenger congestion and reduce the waiting time of passengers.

Next, we also consider the situations with only one type of rolling stocks on the basis of the same train schedule obtained by the proposed methods. For comparative convenience, we consider two operation scenarios with only high-capacity rolling stocks and low-capacity trains, respectively, which are represented by case 4 and case 6 in the following discussion. Then, we compare these two cases with the obtained operation scheme in Fig. 13 (denoted by case 5). Table 14 provides a detailed comparison of the evaluation indexes for different operation schemes, where the meanings of indexes are the same as those in Table 10. Clearly, only

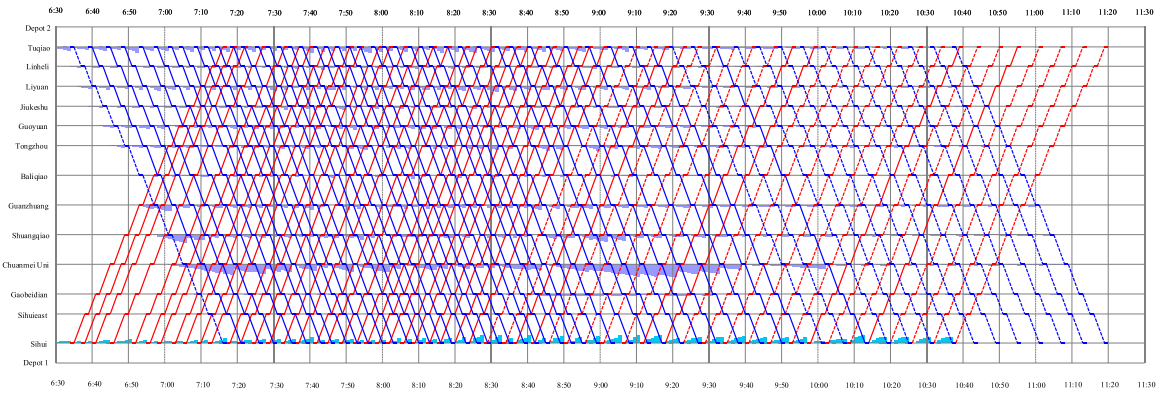


Fig. 14. Passenger accumulation in different station halls without PFCS.

Table 14

Comparison of train operation indexes with different mode.

| Instance | $WT/min$ | $dev^d$ | $CR$   | $CS$  | $C$    | $dev^c$ |
|----------|----------|---------|--------|-------|--------|---------|
| Case 4   | 230642   | –       | 440000 | 62000 | 502000 | –       |
| Case 5   | 231180   | 0.2%    | 367400 | 82000 | 449400 | 10.5%   |
| Case 6   | $\infty$ | –       | 198000 | 62000 | 260000 | –       |

using the high-capacity rolling stocks corresponds to the highest operating costs (i.e.,  $C$ ) and only using the low-capacity rolling stocks corresponds to the lowest operation costs. Since there are more chances that different services can be executed by the same rolling stock when using a single type of rolling stocks, the movement costs for the operation plan in *case 5* are much higher than those of the solutions obtained in *cases 4* and *case 6*. As for the passenger waiting time, it shows a similar phenomenon to that in the small-scale example. Compared with only using the high-capacity rolling stocks (i.e., the operation scheme in *case 4*), the obtained operation scheme in *case 5* can bring 10.5% reduction of the operating costs with only 0.2% increase of passenger waiting time. When we only use the low-capacity rolling stocks, the passenger demand cannot be satisfied entirely. Thus, these results further indicate that the proposed method can simultaneously optimize the train schedule, rolling stock circulation plan and passenger flow control strategy with considering different types of rolling stocks so as to obtain a systematically optimized operation plan for the interests of both passengers and metro companies.

## 6. Conclusions and further works

The temporal and spatial nonequilibrium of passenger flow is a common phenomenon in the metro system. To coordinate the transportation capacity and passenger demand, this paper studied a joint optimization problem of train scheduling, rolling stock circulation planning and passenger flow control, in which different types of rolling stocks with various loading capacities are provided for the operations in different time periods. By mathematically describing the dynamics of passenger demands and the process of trains returning to or exiting from the depots, a mixed-integer nonlinear programming (MINP) model was formulated to minimize the passenger waiting time and the operating costs of the metro system. Through linearization, the MINP model was reformulated as a mixed-integer linear programming model (MILP) that could be solved by a commercial solver (e.g., CPLEX, GUROBI). Then, a heuristic algorithm based on the tabu search algorithm and CPLEX solver was developed to efficiently obtain a near-optimal solution for large-scale instances. Finally, several sets of numerical experiments based on a small-scale example and the Beijing metro Batong line were conducted to demonstrate the effectiveness of the proposed methods. The results confirmed the performance of the proposed methods in saving operating costs and balancing the interests of passengers and metro companies.

Future research can focus on the following three aspects. (1) Since collaboratively optimizing the operation plans of different lines over the whole metro network can help to make full use of the resource (especially the rolling stock), the network operation mode has been regarded as the future mode. Extending this study to a metro network with sharing the rolling stock is an important topic for future work. (2) Due to the dynamic and time-dependent characteristics of passenger flow, historical AFC data do not exactly represent the current passenger flow. Hence, optimizing train operation plans (especially train scheduling) via passenger flow prediction will be investigated in our future research. (3) In this study, the turn-around capacity of each end station is assumed to be large enough, which typically may influence the turn-around operations of different types of trains. In the future, integrated optimization of track allocation, train schedule and rolling stock circulation plan for the applications of operating multiple types of trains will be studied through designing more efficient or exact algorithms.

## CRedit authorship contribution statement

**Housheng Zhou:** Conceptualization, Investigation, Software, Writing – original draft, Writing – review & editing. **Jianguo Qi:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Funding acquisition. **Lixing Yang:** Conceptualization, Supervision, Methodology, Writing – original draft, Writing – review & editing, Funding acquisition. **Jungang Shi:** Supervision, Writing – review & editing, Software. **Pengli Mo:** Software, Writing – review & editing.

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## Appendix. Linearization methods

Some useful properties related to the linearization methods are proposed in the following discussion.

**Property 1** (Bemporad and Morari, 1999). *The product term of two binary variables, e.g.,  $x_1x_2$ , where  $x_1 \in \{0, 1\}$ ,  $x_2 \in \{0, 1\}$ , can be replaced by an auxiliary binary variable  $x_3 = x_1x_2$ , such that  $x_3 = 1 \iff x_1 = 1 \wedge x_2 = 1$ . Then,  $x_3$  should satisfy*

$$\begin{cases} x_3 \leq x_1, \\ x_3 \leq x_2, \\ x_3 \geq x_1 + x_2 - 1, \\ x_3 \in \{0, 1\}. \end{cases} \quad (47)$$

**Property 2.** *The product term of binary variables and continuous variables, e.g.,  $x_4x_5$ , where  $x_4 \in \{0, 1\}$  and  $x_5 \in [0, \mu]$ , can be replaced by an auxiliary continuous variable  $x_6 = x_4x_5$ , such that  $x_4 = 1 \implies x_6 = x_5$ , and  $x_4 = 0 \implies x_6 = 0$ . Then,  $x_6$  should satisfy*

$$\begin{cases} x_6 \leq \mu \cdot x_4, \\ x_6 \leq x_5, \\ x_6 \geq x_5 - \mu \cdot (1 - x_4), \\ x_6 \in [0, \mu]. \end{cases} \quad (48)$$

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